



IMPROVED COOKSTOVE GROUPED PROJECT IN PAPUA NEW GUINEA



Document Prepared by TEM PNG Cookstoves Project Pte Ltd

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PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

The Improved Cookstove Grouped Project in Papua New Guinea by TEM PNG Cookstoves Project Pte. Ltd. (“**TEM**”) is a grouped project activity that involves the distribution of fuel efficient improved cookstoves (ICSs) within the boundary of Papua New Guinea (PNG). The ICS disseminated through this project will replace traditional cooking methods where open-fired wood-fueled cooking is the primary cooking method in the household.

A summary description of the technologies/measures to be implemented by the project:

The technology of the project activity involves the use of the ICS whereby it is designed as a stainless-steel insulated combustion chamber, with a high thermal combustion efficiency (over 25%). This highly effective ICSs uses less biomass fuel and emits less smoke than open-fired wood-fueled cooking methods. The implementation of the ICSs is estimated to reduce the amount of wood cutting, as well as the amount of time spent by end-users foraging for firewood.

Scenario existing prior to the implementation of project activity:

In the baseline scenario prior to the implementation of the project, there would be continued wood cutting practices and the use of non-renewable wood fuel by the target population to meet the thermal energy needs for cooking. With the ICS technology, woody biomass would be more efficiently combusted, hence emitting lower carbon emission and in turn generating GHG emission reductions.

Presently, a total of 19,206 ICSs has been distributed, one to each household, within Phase I of the project. The project and first ICS distributed started on 23 September 2024 and the last ICS distributed for this monitoring period was on 31 October 2024.

The monitoring period set for this monitoring report is **23 September 2024 to 31 December 2024**. All ICSs taken account under the project activity are commissioned. During the monitoring it was found that all of **19,206** devices were operational.

The total GHG emission reduction monitored during the monitoring period is **13,443 tCO₂e**.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation	04/04/2023 – 14/06/2024	Verified Carbon Standard	Earthood Services Private Limited	-
Verification	23/09/2024 – 31/12/2024	Verified Carbon Standard	SustainCERT SA	3 Months 8 Days

1.2.1 Sectoral Scope and Project Type

Sectoral scope ¹	03 – Energy Demand
Project activity type	Type II – Energy efficiency improvement projects

1.3 Project Proponent

Organization name	TEM PNG Cookstoves Project Pte. Ltd.
Contact person	Dr. Gabriel Machovsky Capuska
Title	Director of International Projects
Address	15th Floor, 31 Market Street, Sydney, NSW 2000, Australia
Telephone	+61 3 5977 1286
Email	gabriel@tem.com.au

¹ Projects, activities, or methodologies may be developed under any of the 16 VCS sectoral scopes: <https://verra.org/programs/verified-carbon-standard/vcs-program-details/#sectoral-scopes>

1.4 Other Entities Involved in the Project

TEM PNG Cookstoves Project Pte. Ltd. is the sole entity involved in the project and no other entities were involved in the project activity, therefore this section is not applicable.

Organization name	Not applicable
Role in the project	
Contact person	
Title	
Address	
Telephone	
Email	

1.5 Project Start Date

Project start date	23 September 2024
Justification	First End-User Agreement signed with end-user.

1.6 Project Crediting Period

Crediting period	☐ Seven years, twice renewable ☒ Ten years, fixed ☐ Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)
Start and end date of first or fixed crediting period	23 September 2024 to 22 September 2034

1.7 Project Location

The GPA will distribute ICSs to households within the project boundary of PNG (Latitude: 6.3150° S Longitude: 143.9555° E). The first project instance monitored in this monitoring report is across the 10 local level governments (LLGs) in 4 districts of the Southern Highlands Province. Map of PNG and specific coordinates of the province are shown in Figure 1 and Table 1 below:

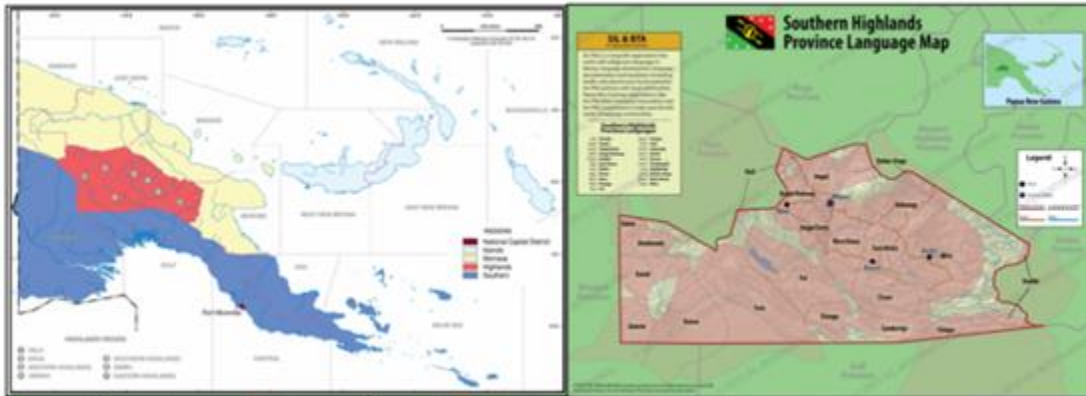


Figure 1: Map of Papua New Guinea including the Highlands Region (left) and map of Southern Highlands Province (right). Maps extracted from www.pnglanguages.sil.org

Table 1: The Geographic coordinates of the province involved as the first project instances of the group activity in Papua New Guinea

S/N	LLGs	District	Province	Latitude	Longitude
1.	Karints	Mendi/Munihu	Southern Highlands	6.4179° S	143.5636° E
2.	Upper Mendi				
3.	Mendi Urban				
4.	Kagua	Kagua/Erave			
5.	Lower Mendi	Imbonggu			
6.	Imbonggu				
7.	Ialibu Basin				
8.	Kewabi	Ialibu/Pangia			
9.	East Pangia				
10.	Ialibu Urban				

1.8 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0050	VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves	1.0
Standard	-	Sampling and surveys for CDM project activities and programmes of activities	9.0
Tool	30	Calculation of the fraction of non-renewable biomass	4.0
Guidelines	-	Sampling and surveys for CDM project activities and programmes of activities	4.0

1.9 Double Counting and Participation under Other GHG Programs

1.9.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.9.2 Registration in Other GHG Programs

Was the project registered or seeking registration under any other GHG programs?

☐ Yes ☒ No

1.10 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.10.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

1.10.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

1.10.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

Response:

The project activity involves the distribution of ICSs to households within PNG. The boundary of the emission reductions is limited to the decrease in firewood consumption of each household, hence there are no indirect upstream or downstream GHG emissions to the organization's supply chain.

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through the Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☒ No

1.11 Sustainable Development Contributions

Sustainable Development Contributions of Project Activities

The project will enable and enhance households to achieve several Sustainable Development Indicators (SDI):

The project provides ICSs with high thermal efficiency to reduce the consumption of firewood which helps achieve efficient use of an important type of nature resources, non-renewable biomass. This impacts the estimated reduction in removal of woody biomass to the tune of 3.70 tonnes of firewood per ICS per year, from forests surrounding the communities thereby leading to an increase in above-ground biomass in these forests. This value is estimated from the difference between the amount fuel used with open-fire cooking ($BC_{b,i,y}$) and the amount of fuel used with the ICS ($BC_{p,j,k,y}$) based on the VM0050 Methodology. (SDI Goal 15.2.1).

The project activity will replace 100,000 traditional cookstoves with fuel efficient ICSs. The benefits of changing from a traditional to a more efficient ICS results in the consumption of lesser firewood that is usually purchased at high prices and cleaner cooking. Hence, this reduces the economic burden of families under the project activity (SDI Goal 1) and ensures households have access to clean cooking (SDI Goal 7).

The ICS also reduces the levels of fine particulate matter (PM_{2.5}) emissions in the air compared to the baseline emission level of 943.30 mg/MJ, thus, improving the health and well-being of the end-users involved in the project. (SDI Goal 3)

With the project activity, jobs are generated for house registration and ICSs distribution for participants. Also, additional jobs for the local community during sampling for verification, for the maintenance of ICSs and various project activity will be required throughout the project lifetime. This creates long term jobs for the local community, bringing opportunities and inclusiveness to people from different genders, age, occupation, and persons with disabilities. TEM envisions that 50% of the project's employees will be female. (SDI Goal 8 and SDI8.5.1)

With the creation of jobs, the project proponent will also carry out training for vocational and relevant skills of local individuals by providing non-formal education and training on issues related to climate change, with specific skill building in operations and surveying activities related to ICS distribution and its monitoring under VCS. (SDI Goal 4.1 and SDI Goal 4.2)

Lastly, there is a contribution to the GHG emission reduction through an estimated reduction of ~3.20 tCO_{2e} per ICS due to the replacement of baseline stoves with ICSs over the crediting period. (SDG 13).

Sustainable Development Contributions to nationally stated sustainable development priorities

The UN Sustainable Development goals were adopted by all member countries of the United Nations in September 2015. These goals are closely aligned with the 'PNG Vision 2050' and the National Strategy for Sustainable Development known as 'StaRS' which have now become the focus of sustainable development activities in PNG². The PNG Vision 2050 outlines the need for economic growth in local communities through skill development, reduction of greenhouse emissions and conservation of biodiversity which is aligned with the SDGs stated in Table 2. For StaRS, this project aligns with the strategy advocated for sustainable economic growth while preserving the environment around.

Finally, this project also aligns perfectly with the PNG National Health Plan for 2021-2030. The PNG Government's vision is for a healthy and prosperous nation where health and wellbeing are enjoyed by all. The goal is to prevent ill health, identify and address health risks and emerging diseases, and provide accessible and affordable quality healthcare for all. This is aligned with the project activities which causes a reduction in the level of fine particulate matter (PM_{2.5})

² PNGDNPM. (2021). *Papua New Guinea Vision 2050* | Papua New Guinea Environment Data Portal. <https://png-data.sprep.org/dataset/papua-new-guinea-vision-2050>

emissions within households as stated in Table 2, Row 2, which addresses the health risks related to fine particulate matter.

Table 2: Sustainable Development Contributions

Row number	SDG target	SDG indicator	Net impact on SDG indicator	Current project contributions	Contributions over project lifetime
1)	1.1	Goal 1: Amount of money saved by families that buy firewood	Implement activities to increase	For this current monitoring period, a total of 19,206 traditional cookstoves are replaced with fuel efficient ICSs in the project area, with a focus in the Southern Highlands Province. It is estimated that the savings per household per day would be 1.04³ Kina in households purchasing firewood. By proportioning the number of households that purchase their firewood from the monitoring survey, an estimated 57,890 Kina⁴ is saved across 1000 households (5.2%) in this monitoring period.	The efficiency of changing from a traditional cookstove to a more efficient one will result in the consumption of lesser firewood that is usually purchased at high prices,⁵ hence reducing the economic burden of families that buy firewood under the project activity. The project activity has replaced 19,206 traditional cookstoves with fuel efficient ICSs, leading to an estimated savings of 1.04 Kina per household per day. By proportioning the number of households that purchase their firewood from the monitoring survey, an estimated 57,890 Kina⁶ is saved across 1000 households (5.2%) in this monitoring period.

³ The calculation method is demonstrated in the SD VISTa Monitoring Plan Spreadsheet made available to validators.

⁴ The calculation method is demonstrated in the SD VISTa Monitoring Plan Spreadsheet made available to validators.

⁵ ACIAR 2013_Promoting diverse fuelwood production systems in Papua New Guinea

⁶ The calculation method is demonstrated in the SD VISTa Monitoring Plan Spreadsheet made available to validators.

2)	3.9	Goal 3: Reduction of fine particulate matter (PM _{2.5}) emissions within households to reduce air pollution within households in the project area.	Implemented activities to increase	Emissions of PM_{2.5} were reduced to be below the baseline emission level by 81% to 177 mg/MJ⁷ in project households for newly distributed ICSs. For ICSs distributed for more than a year, there would be a decrease in thermal efficiency, however, for this current monitoring period, no ICSs were more than a year old, as all 19,206 ICSs were newly distributed. Overall, this resulted in a total decrease of 244 tonnes of PM_{2.5}⁸ during this monitoring period. Additionally, this is further supported by the Ground Monitoring Survey which reported a 100% of households have observed a decrease in intensity of the smoke compared to the baseline scenario.	Contribute to improved health and well-being brought about by reduced levels of fine particulate matter (PM_{2.5}) emissions within households below the baseline emission level of 943 mg/MJ delivered⁹. Overall, a total decrease of 244 tonnes of PM_{2.5}¹⁰ has been observed during the project lifetime.
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⁷ http://catalog.cleancookstoves.org/stove_details.html?stove_id=stove_AW0UJ

⁸ The calculation method is demonstrated in the SD VISTa Monitoring Plan Spreadsheet made available to validators.

⁹ http://catalog.cleancookstoves.org/stove_details.html?stove_id=stove_A8WGMJX

¹⁰ The calculation method is demonstrated in the SD VISTa Monitoring Plan Spreadsheet made available to validators.

3)	4.3	Goal 4.1: Number of youth and adults in formal and non-formal education and training in the previous 12 months, by sex	Increase	Non-formal education in the form of vocational and relevant skills was provided to 24 local individuals with respect to issues related to climate change, with specific skill building in operations and surveying activities related to ICS distribution and its monitoring under VCS have been provided during this monitoring period. All staff met the minimum passing rate of the quizzes given to them.	Contributed to increasing vocational and relevant skills of local individuals by providing non-formal education and training on issues related to climate change, with specific skill building in operations and surveying activities related to ICS distribution and its monitoring under VCS. During the project lifetime, a total of 24 local staff were trained in training courses. These training courses were conducted annually for all the staff to make sure that the knowledge transferred is up to date and the delivery of the project remains of a high standard. At the end of each training session, staff sat for an informal quiz to test their understanding of the training material. The measurement of improvement of learning outcomes and skill development will be based on the passing rate of the quiz. The minimum passing rate is at least 70%, and if a staff falls below the mark, they will have to re-sit the quiz within the same day.
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4)	4.4	Goal 4.2: Number of youth and adults with technical and vocational skills, by type of skill within the project area.	Increase	During this monitoring period, a total of 24 local staff were trained to conduct regular monitoring of the project sites for additional monitoring and maintenance of the ICSs and project activity. This is alongside training in how to use the ICS, its advantages, as well as how it contributes to climate change and the carbon cycle. The local staff will be annually trained to make sure that the knowledge transferred is up to date and the delivery of the project remains on high standards.	Contributed to increasing the number of youth and adults who have relevant skills, including technical and vocational skills, for employment, decent jobs, and entrepreneurship. An estimated 24 local staff and personnel were trained during the project lifetime to conduct regular monitoring of the project sites for additional monitoring and maintenance of the ICSs and project activity. This is alongside training in how to use the ICS, its advantages, as well as how it contributes to climate change and the carbon cycle. The local staff will be annually trained to make sure that the knowledge transferred is up to date and the delivery of the project remains of a high standard.
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5)	5.4	5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age, and location	Implemented activities to decrease	During this monitoring period, the proportion of time spent cutting, collecting and carrying firewood from trees far away from households was monitored to be 1 time / week for people staying in the project area. This decrease from the baseline scenario results in a total decrease of 762,759 trips across all participants during the monitoring period.	The project will reduce women and children's drudgery through time savings in reducing time spent on cutting, collecting, and carrying firewood from trees far away from households. Monitored at 3-5 times/week in the baseline scenario as per Tasman Environmental Market's Baseline Survey Report, participating households were monitored to only collect firewood at a frequency of 1 time/week. As such, this decrease from the baseline scenario results in a total decrease of 762,759 trips across all participants during the project lifetime.
6)	5.5	Goal 5: Number of women in the project including managerial roles	Implemented activities to increase	Out of the total 24 local individuals that has been trained and hired for this project, 9 (37%) are female.	Ensure women's full and effective participation and equal opportunities for leadership at all levels of decision-making in the project. During the lifetime of the project activity, a total of 24 local individuals that has been trained and hired for this project, 9 (37%) are female. Monitored as the fair and unbiased employment of women within the local team during the lifetime of the project. The project envisages that 50% of its employees will be female.

7)	7.1	Goal 7: Number of people with primary reliance on clean fuels and technology	Increase	As of this current monitoring period, a total of 19,206 traditional cookstoves are replaced with fuel efficient ICSs in the Southern Highlands project area.	The project contributes to increasing access to ICS technology with Burn cookstove installations in 19,206 households that have been using traditional three-stone fire, over the project lifetime.
8)	8.3	Goal 8: Number of people formally employed, by sector and sex under the project activity.	Implement activities to increase.	During this current monitoring period, a total of 2 full-time and 22 part-time local staff were hired. Out of the 24 staff, a total of 9 (37%) are females. A ratio of male to female employees is 5:3 All the staff will be annually trained to make sure that the knowledge transferred is up to date and the delivery of the project remains on high standards.	With the inception of the project activity, jobs will be created firstly for the distribution of ICSs. Personnel will also be trained in how to use the ICS, its advantages, as well as how it contributes to climate change and the carbon cycle. Further to that and aside from the monitoring done during the verification cycle, the project activity will also conduct regular monitoring of the project sites for additional monitoring and maintenance of the ICSs and project activity. In turn creating long term jobs for the local community, without prejudice of sex, age, occupation and persons with disabilities. During the lifetime of the project activity, a total of 2 full-time and 22 part-time local staff were hired. Out of the 24 staff, a total of 37% are females. All the staff will be annually trained to make sure that the knowledge transferred is up to date and the delivery of the project remains of a high standard.

9)	8.5	8.5.1 Average hourly earnings of employees, by sex, age, occupation, and persons with disabilities	Implement activities to increase.	In this monitoring period, out of the total of 24 staff, 22 personnel were employed part-time. These part-time employees were compensated with an average income of (K 1,343 / month) and is pro-rated where part-time employment is relevant. This value considers age, sex, occupation and persons with disabilities.	Contribute to an increased economic income by providing fair compensation per month for the employees performing the jobs created as part of the project activity initially during deployment and later during project monitoring. Compensation will differ depending on the job scopes assigned to each formal employee employed under the project activity and will be sampled based on this distinction. Individuals employed under the project activity will likely be community members who are currently unemployed or working in the informal sector with an average income estimated as K439/month (Bonilla, et al., 2020). During the lifetime of the project activity, a total of 24 staff, 22 personnel were employed part-time. These part-time employees were compensated with an average income of (K 1,343 / month) and is pro-rated where part-time employment is relevant.
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10)	13.0	Tonnes of greenhouse gas emissions avoided or removed	Increase	Total emission reduction in the current monitoring period is 13,585 tCO ₂ eq.	Contribute to GHG emission reduction through an estimated reduction of ~3.20 tCO₂e per ICS due to the replacement of baseline stoves with Burn ICSs over the crediting period. Over the lifetime of the project activity, a total of 19,206 ICSs were distributed, generating a total of 13,443 tCO₂ eq.
11)	15.2	15.2.1 Progress towards sustainable forest management through the increase of above-ground biomass in forests	Implemented activities to increase	The project has resulted in saving 3.70 tonnes of non-renewable biomass per ICS per year over this monitoring period leading to total savings of 13,525 tonnes of biomass from 19,206 ICS installations.	Contribute an estimated reduction in removal of woody biomass to the tune of 3.70 tonnes of firewood per ICS per year, from forests surrounding the communities thereby leading to an increase in above-ground biomass in these forests. The above value is based on the difference between the amount fuel used with open-fire cooking (BC_{b,i,y}) and the amount of fuel used with the ICS (BC_{p,j,k,y}) based on the VM0050 Methodology. Over the lifetime of the project activity, a total savings of 13,446 tonnes of biomass from 19,206 ICS installations.

1.12 Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	Under this project, the stakeholders are identified as local people, communities and/or representatives who are directly or indirectly affected by the activities proposed, Examples of identified stakeholders are the end-users and ICS manufacturers. Aligned with the United Nations Convention on the Rights of Indigenous People (2007)¹¹ and the United Nations Development Program (UNDP) Stakeholder Engagement Guidelines (2022)¹², the development of a project encompasses engagement with multiple stakeholders at different stages that spans from Project Identification to Implementation.
Legal or customary tenure/access rights	The project aligns with several International and Papua New Guinean's laws, regulations and conventions. TEM, as the project developer, and its in-country team will follow and obey all these relevant laws, regulations and conventions (as stated below): <u>International Laws, Regulations and Conventions:</u> • United Nations Framework Convention on Climate Change (UNFCCC,1992) • United Nations Declaration on the Rights of Indigenous Peoples Rights (2007)

¹¹ <https://www.ohchr.org/sites/default/files/Documents/Issues/IPeoples/UNDRIPManualForNHRIs.pdf>

¹² https://ses-toolkit.info.undp.org/sites/g/files/zskgke446/files/SES%20Document%20Library/Uploaded%20October%202016/UNDP%20SES%20Stakeholder%20Engagement%20GN_Final_rev_July2022.pdf

	• Stockholm Declaration of the United Nations Conference in Human Environment (1972) • Rio Declaration on Environment and Development (1992) • Paris Agreement (2015) <u>Papua New Guinea's Laws, Regulations and Conventions:</u> • Papua New Guinea's Constitution (1975) • Climate Change Management Act 2015, Amended (2022), and proposed Carbon Market draft regulation (2023) • Forestry Act (1991) • Environmental Act (2000) • Investment Promotion Act (1992) • Industrial Relations Act (1962) • Industrial Health Safety and Welfare Act (1962) • Employment Act No. 54 of 1978 (Consolidated up to 31 March 2001) • Child Welfare Act (1961) • Discriminatory Practices Act (1963) • Workers Compensation Act (1978) • Papua New Guinea Vision (2050)
Stakeholder diversity and changes over time	<u>Social Capita:</u> Papua New Guinea (PNG) is characterised by a high degree of income inequality with 40% of people living below the extreme poverty line and 41% of children living in poverty¹³. Lack of education, healthcare and infrastructure are known as the main contributors to influence poverty in the country. In addition, 80%¹⁴ of the population lives in rural areas without the benefits of electrification, vaccinations, or basic ICSs. It is estimated that the country's rural population increased 1% from 2021 to the 8,765,048 measured in 2022¹⁵.

¹³ <https://data.unicef.org/how-many/how-many-children-live-in-poverty/>

¹⁴ https://www.mypsup.org/countries/Papua_New_Guinea#:~:text=Some%2080%20per%20cent%20of,country%20has%20several%20urban%20centres.

¹⁵ <https://www.macrotrends.net/global-metrics/countries/PNG/papua-new-guinea/rural-population#:~:text=Papua%20New%20Guinea%20rural%20population%20for%202022%20was%208%2C765%2C048%2C%20a,%202.06%25%20increase%20from%202019.>

Health and Well Being:

In accordance with the WHO's guidelines, the air quality in Papua New Guinea is considered moderately unsafe, with the annual mean concentration of PM_{2.5} exceeding the recommended maximum of 10 µg/m³^{16,17}. Vehicle emissions, the mining industry, the oil and gas industry, and waste burning are suggested as the main drivers for this pattern. This is also magnified by indoor air pollution and smoke generated from the burning of solid fuels by traditional open fire cooking and heating that is linked to various illnesses and deaths¹⁸.

The Baseline Study (file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf) showed that open fire cooking (i.e. inefficient three-stone fire burning non-renewable woody biomass) produced thick grey or light blue smoke that led to headaches, cough, asthma, and other related diseases.

Economic Capita:

The population in PNG's is largely rural and growing with around 60% of the population being of working age (15-64 years) as shown in Figure 2

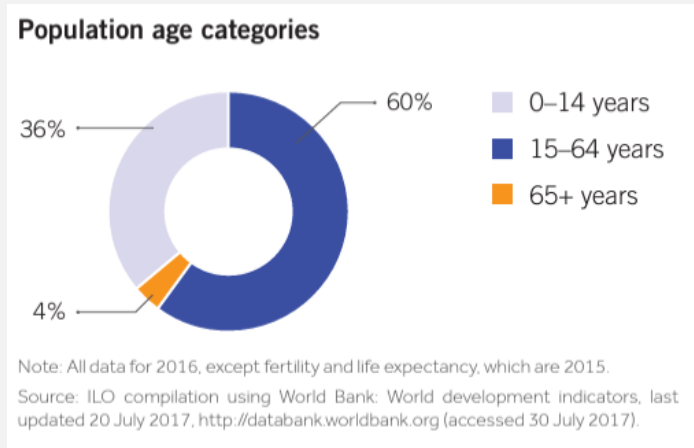


Figure 2: Population age categories, depicting age groups that are too young to work (0-14 years), within a typical working age (15-64 years), and retired (65+ years).

According to Employment and Environmental Sustainability Fact Sheet (2017), the PNG labour force participation rate was 70%, and the employment-to-population ratio was 68%, both close to gender parity. According to this study, the vulnerable employment in the country accounts for 44% of the labour force, with many of

¹⁶ World Health Organization (WHO), Health and environment scorecard: Papua New Guinea, Air Pollution

¹⁷ [https://www.who.int/en/news-room/fact-sheets/detail/ambient-\(outdoor\)-air-quality-and-health](https://www.who.int/en/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health)

¹⁸ World Health Organization (WHO), Indoor smoke from solid fuels

those workers having own-account status. The split between Vulnerable employment, Employees, and Employers is shown in Figure 3 below:

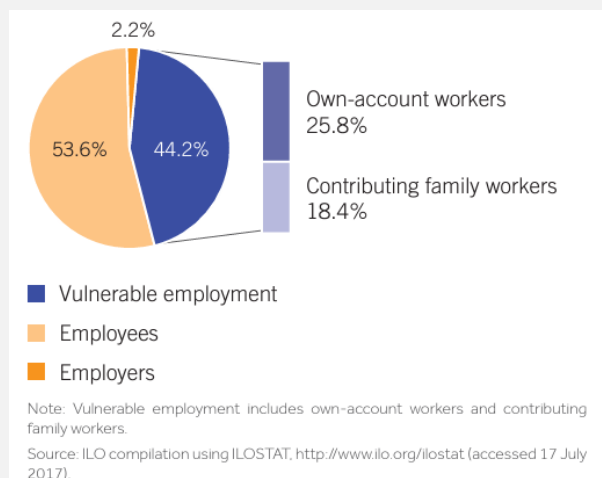


Figure 3: Presentation of Vulnerable Employment in Papua New Guinea by status in 2017

Own-account and contributing family workers are more likely to experience low job and income security than employees and employers, as well as lower coverage by social protection systems and employment regulation.

Informal sector employment is particularly widespread among the youth. A gender gap in access to formal sector employment means that men are more likely to find formal jobs than women. Wages in the informal economy are considerably lower than in the formal economy¹⁹. This pattern is particularly relevant to the rural areas of the Highlands region that is characterized by low and informal income, mostly derived from the sales of agricultural produce (e.g., fresh food, coffee and firewood) in street markets (Figure 4), with 40% of the population living near the poverty line (Hanson, et al., 2001²⁰), with an average income estimated as K439/month (Bonilla, et al., 2020²¹).

Expected changes in well-being

The distribution of ICSs within the project boundary is expected to reduce the amount of firewood used for cooking and heating

¹⁹ Improved Technical and Vocational Education and Training for Employment Project: Report and Recommendation of the President, <https://www.adb.org/projects/documents/png-53083-001-rrp>

²⁰ Hanson, L., Allen, B., Bourke, R. (2001). Papua New Guinea Rural Development Handbook. Canberra, Australian National University

²¹ Bonilla, F., Voorend, K., Anker, R., Anker, M., & Prates, I. (2020). Anker Living Income Reference Value: Rural Papua New Guinea (2020) (No.20-03-20). Universidad Privada Boliviana.

	by the end-users. It is also expected that the local communities will incur less trips per week to collect firewood. In those households where firewood is purchased, an increase in fuel efficiency will result in monetary savings during the project lifetime. The continued usage of the ICSs by local communities will also gradually reduce their exposure to high CO and PM2.5 and due to higher combustion efficiency of this new technology over the lifetime of the project activity. The project aims to hire around 130 local people through a fair and unbiased process, envisioning 50% gender equality over the project lifespan. During the monitoring period, a total of 24 local individuals had been trained and hired for this project, of which 9 (37%) of them were female.
Location of stakeholders	The location of stakeholders for this monitoring period are within the project boundary of PNG, within the Southern and Western Highlands Province as per Section 1.8.
Location of resources	The project activity involves the distribution of ICSs to individual households and does not consider any ownership over an end-user's territory, therefore this section is not applicable to the project activity.

2.1.2 Stakeholder Consultation and Ongoing Communication

During this monitoring period, stakeholder consultations presented are focused on Southern Highland Province where distribution and monitoring of ICSs took place.

Date of stakeholder consultation	Southern Highlands Province: 13 March 2023
Stakeholder engagement process	Prior to the LSCs, TEM utilized several media outlets to advertise the event to the provinces. These efforts were fulfilled using: • Newspaper Advertisements • Letters of Notification to Provincial Governments • Radio Broadcasts • Invitation Letters to Churches • Social Media Posting • Webinar

- Email to PNG Australia Alumni Association (PNGAAA) members

All advertising outlets were run in a staggered format with the aim of keeping stakeholders informed and aware of the LSCs that would be happening within their provinces. For the exact dates of advertisements being published, please refer to the PD Appendix II: LSCs Outreach.

The stakeholders engaged in this project were people or groups of people that were directly or indirectly affected by the project as well as those who were interested in the project and or had the ability to influence its outcome either positively or negatively. They include:

- Representatives from the provincial administration
- Representatives from Women Groups
- Representatives from Churches
- Ward councilors
- Heads of households both male and female
- Key informants such as teachers, community health workers etc.
- ICS (SHP only)
- Youth leaders and
- Educated elites (lecturers, nursing officers, Health Extension Officers, Police personals etc.)

Southern Highlands Province LSC

A total of 159 people attended this LSC on the 13th of March 2023 held at Kumin Catholic Mission in Mendi, SHP. Including those who commuted into Mendi from various districts, there were public servants such as nursing officers, police, teachers and church elders and pastors present during the consultation.



Figure 4: A demonstration by Ms. Magret Yangpin and Mr. Richard Manda – SHP Field Assistants.

The LSC was facilitated by Mr. Michael Kisombo, the project Manager with Simi Oa Consultancy Services who was engaged by TEM PNG, did a power point presentation after welcoming and acknowledging all the stakeholders that were present. The PowerPoint slides were prepared in English, and explained in tok pidgin, the common language where most could understand well as those participated came from different local dialects.

Attendees were briefed on the purpose of the meeting, and informed of the:

- Carbon cycle and the important role it played within their forests.
- Findings of the Baseline Study conducted in the Highlands region including SHP and WHP (file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf)
- The Verra registration process
- The role of carbon finance in the project activity
- Indicative timeline and implementation plan of the project

Two field assistants as seen in Figure 4 were then invited to demonstrate and to explain how the ICS is used using the sample that was exhibited during the LSC. That was further reinforced by one of the users who was also invited to give her experience of how she is currently using the ICS and explained the benefits her family is experiencing.

After the presentation, stakeholders were given the opportunity to ask questions and provide their reflections. For the summary of questions asked and subsequent

	answers provided by TEM PNG, please refer to Appendix III: Stakeholder Feedback. After the discussions, there were about 29 feedback forms that were received through the feedback forms. These feedback forms may be made available to validators upon request. At the end of the LSCs, stakeholders were informed of the project's email (pngcookstoves@tem.com.au), as well as direct contact details to the local project manager and lead project manager in TEM PNG. In all marketing documents, the project email and website were also published, to ensure stakeholders had multiple avenues to reach out to TEM PNG with feedback, allowing for on-going communication.
Consultation outcome	There was no negative feedback received during the LSC. Most questions aimed to seek clarity on how a more efficient ICS could reduce carbon in the atmosphere, when the project could begin, and if more ICSs could be distributed. For a complete list of questions and feedback, please refer to the Appendix III: Stakeholder Feedback. For more information, a Local Stakeholder Consultation Report is available upon request of the validator.
Ongoing communication	During the Local Stakeholder Consultations, it was discussed how stakeholders may keep in contact with TEM's in-country project manager, Michael Kisombo. During which, the following solutions were offered: 1. Reaching out to the in-country Team during the twice a year visits to maintain the ICSs. 2. Reaching out to TEM through the project's dedicated email. 3. Reaching out to TEM through the project's website. The following website of the project activity was available to the stakeholders: https://www.tasmanianenvironmental.com.au/pngcookstoves/ The following email addresses will be also used for inquiry or comment on the project: pngcookstoves@tem.com.au TEM PNG Country Manager: Michael Kisombo, michael.kisombo@tem.com.au. TEM's Office in Papua New Guinea (TEM PNG): +675 71532292
Stakeholder input	Please refer to Appendix III: Stakeholder Feedback.

Overall, TEM engaged with ICS end-users during visits to register the households and to give out the ICSs. In total, TEM has registered and distributed a total of 19,206 ICSs during this monitoring period. In addition, monitoring surveys and training of new staff was also conducted.

For more information on stakeholder consultations conducted during the monitoring period, please refer to Appendix IV: Stakeholder Consultations.

2.1.3 Free, Prior, and Informed Consent

Obtaining Consent	The project is voluntarily implemented by TEM PNG Cookstoves Project Pte. Ltd., and end-users are free to choose whether they will be involved in the project or not. Free, prior, and informed consent takes place during household registration and ICS distribution through signing off an End User Agreement that they would like to be part of the project, have received the ICS, and they will transfer the rights of the carbon credits generated from the project to TEM PNG Cookstoves Project Pte. Ltd. End-users are free to withdraw from the project activity at any point in the project lifetime.
Outcome of FPIC	For end-users to be considered part of the project, their consent will be sought at two major points of the project onboarding process: 1. Household Registration 2. ICS Distribution During the household registration phase, the end-user's consent to participate in the project is obtained. Once the willingness to participate is acknowledged, information on the house geographical location and alongside a list of prerequisite questions relating to their baseline cooking behaviors are collected. The end-user is then made aware of what the project entails, how to collect their ICS, and who they may contact if they have any questions before they <u>sign off</u> on a Household Registration Form, marking their consent to join the project activity. They are then given a Token Flyer that contains an individual code that is linked to their house and name, which they will bring down to the ICS distribution site to collect their ICS. Field assistants will take a picture of the end-user with their token flyer and individual to be stored in our database. During the ICS distribution phase, the end-user presents the token flyer to a field assistant who checks the token flyer's code against TEM's database to match the end-user's face to a picture within the system. Before the ICS is dispensed at no cost to the end-user, the field assistant briefs the end-user on how to use and maintain the ICS and the environmental, health economic benefits of using the ICS. Once that is acknowledged, the field assistants explain the EUA stating that the carbon offsets generated by that individual's ICS will be transferred to TEM. Upon the end-user's consent, the EUA is signed, and the end-user is given the ICS at no cost.

As the project activity involves the distribution of ICSs to individual households, no land has been encroached or has there been any relocation of people without consent nor forced physical or economic displacement.

2.1.4 Grievance Redress Procedure

As no grievances were recorded during the monitoring period, this section is not applicable.

The details on procedure for feedback and grievance have been made publicly available to all stakeholders on the project website at <https://www.tasmanenvironmental.com.au/pngcookstoves/>

Additionally, a Grievance Redress Mechanism (GRM) Standard Operating Procedure (SOP) has been prepared by TEM PNG and aims to present a set of guidelines for designing and implementing different mechanisms that can address a variety of grievances arising when installing the ICS in PNG during the crediting period of the project activity. The guidelines are prompt and effective and are timely to inform the complainants of the actions taken to resolve issues raised.

The development process and grievance redress procedure is described below:

Development process

The Grievance Redress Mechanism (GRM) is a community driven approach built on traditional Melanesian customary protocols, that are formalised to accept, assess, and resolve feedback or complaints regarding the implementation of the project.

The different channels to report grievances are summarised below in **Table 3**, whereas the process of organising common types of grievances is captured in **Table 4**. The Procedure recognises and prioritises the traditional and customary means of conflict resolution in Melanesian society by encouraging dialogue through clan leaders and staff at the community level, and utilising Field Assistants who are sourced from local communities to act as the first responders while addressing grievances. Outcomes of the GRM are recorded in the GRM logbook. These measures are taken to enhance Project accountability and transparency and to support the project initiatives.

Grievance redress procedure	Table 3: Channel for Grievance Reporting		
	Grievances Reporting Channel	Grievance Mechanism	Reporting
	Physical walking in	Complaints to TEM PNG Office and/or its staff	
	Social media and online	Complaints made online to TEM PNG Office, and/or in country Project manager	
	Phone call	Phone calls to Field Assistance, and/or Project manager	
	Verbal through community representatives	Complaints through community representatives who then channel through the organizational structure	
	Table 4: Process of organising common grievances		
	Type of Grievance	Scope	How will be facilitated
	Unfair distribution of ICS	Cookstoves unfairly distributed by project staff	Gather information, investigate and refer to project lead
	Burglary, sold or destroyed by natural disaster	Cookstoves were stolen, destroyed, or sold	Gather information, investigate, and refer to project lead
	Household missing out on the distribution	Cookstoves missed targeted households	Investigate and refer to project lead for possible re-issue
	Damaged cookstoves	Cookstoves issued were damaged or have technical issues	Investigate and refer to project lead for possible replacement
	Complaints raised by project staff	Unfair decisions related to pay or disciplinary actions	Gather information and refer to project lead

With reference to the GRM, all complaints and grievances will be registered and monitored by the in-country Project Manager supported by the field assistants. These complaints and grievances will be resolved using traditional and customary ways.

This document will be socialized and publicized to communities throughout the project location in their local language, Pidgin English.

2.1.5 Public Comments

No public comments were received during the monitoring period, and none were received outside of the public comment period; therefore, this section is not applicable.

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

TEM has been developing carbon projects since 2015 and has expertise in managing and registering carbon projects, offering its clients comprehensive end-to-end solutions to traversing the carbon market landscape. TEM has 6 registered Australian Carbon Credit Unit (ACCU) registered projects.

Where additional expertise is required for the project activity, TEM consults with in-country experts and universities to acquire a clearer understanding of the socio-economic situation on the field within which the project activity is residing.

2.2.2 Risk assessment

	Risk identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing	No risks identified	The wellbeing of the end-users is improved by less smoke and more safe cooking devices compared to the baseline cooking practice over three-stone fire.
Risks to stakeholder participation	Low adoption or participation rates of end-users in the project	Participation in the project is voluntary, and no household is forced. Outreach and engagement to educate the end users on the direct and indirect benefits of ICSs over three stone fire, and continuous awareness campaigns to boost adoption rates.
Working conditions	Safe working conditions for support staff employed in awareness and distribution of ICSs	Support of local police, private security teams and the involvement of community leaders in ICS distribution planning enables the mitigation of identified security risk for on-ground teams.

Safety of women and girls	No risk identified	The project activity increases women's and girl's safety by reducing the number of trips to the forest to collect firewood through the adoption of ICSs that are safer than open fire practices.
Safety of minority and marginalized groups, including children	No risk identified	The ICS distribution and monitoring do not affect the safety of minority and marginalized groups. The ICSs reduce the risks associated with open fire cooking practices for children.
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	End-of-life waste from ICSs disposal	The ICS is made of stainless-steel parts that are recyclable and do not contribute to hazardous waste. In addition, the project activity decreases the risk of pollutants (smoke from cooking) and does not produce any form of pollution.

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

In accordance with PNG's laws and regulations, there are acts that has been adopted in this project to ensure there is respect for human rights and equity. Some examples that are followed are The Employment Act No. 54 of 1978, The Child Welfare Act 1961, The Discriminatory Practices Act 1963, and The Workers Compensation Act 1978. These laws are further described in our PDD.

In this project, any form of discrimination and harassment is opposed fully and will not be tolerated. Additionally, any forced labour, human trafficking or child labour is opposed in this project and serious action will be taken to the parties involved.

This is stated in our TEM Employee Handbook and Modern Slavery Policy:

	Risks identified ²²	Mitigation or preventative measure(s) taken
Discrimination	Unfair treatment due to a certain attribute such as a worker's racial or ethnic background, political or religious beliefs, stereotypes, age, race or disability.	Under the 'Discrimination' section of the TEM Employee Handbook, all discrimination faced is reportable and action will be taken against the perpetrator. Severe or repeated breaches can lead to formal discipline up to and including dismissal.
Sexual harassment	Sexual Harassment may affect a person mentally and result in a negative change in behaviour.	Under Section 13 of the TEM Employee Handbook, all sexual harassment faced is reportable and the policy provides the appropriate personnel to contact in the event of a sexual harassment. Serious actions will be taken against the perpetrator if found guilty.
Gender equity in labor and work	Workers being treated unfairly due to gender, causing a divide in the workplace.	Workers will be paid fairly based on only their qualifications and their merits.
Forced labor	Workers being treated unfairly and working involuntarily.	Under TEM's Modern Slavery Policy, TEM is dedicated to preventing modern slavery in all its forms, including forced labour, human trafficking, child labour, and debt bondage.
Child labor	Children under 18 are employed, which affects their education and mental and physical health.	Under TEM's Modern Slavery Policy, TEM is dedicated to preventing modern slavery in all its forms, including forced labour, human trafficking, child labour, and debt bondage.
Human trafficking	People being transported around to benefit from their work or	Under TEM's Modern Slavery Policy, TEM is dedicated to preventing modern slavery in all

²² The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

	service as a form of forced labour.	its forms, including forced labour, human trafficking, child labour, and debt bondage.
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2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken
No Risk Identified	This project is a voluntary project activity that involves the distribution of ICSs to individual households free of charge.

2.3.3 Indigenous Peoples and Cultural Heritage

Local people from the project area have been employed by TEM PNG Cookstoves Project Pte. Ltd to run the day-to-day operations of the project activity.

The project is aligned and strictly complies with international laws and regulations such as the UN Declaration on the Rights of Indigenous Peoples. Any form of opposition to these laws will not be tolerated. Risks identified were based on a lack of knowledge on Indigenous Peoples and Cultural Heritage in the project area and with their respective preventive measures have been included in the table below:

Risks identified	Mitigation(s) or preventative measure taken
Lack of knowledge of the rights of the Indigenous Peoples	Under the regulatory surplus of the project activity, the UN Declaration on the Rights of Indigenous Peoples Rights states that this project complies with the aims of the UN Declaration and respects the rights of the Indigenous people. Personnel will be well educated on the rights of the Indigenous people. Any issues or problems observed can be brought up as a Grievance as explained more in section 2.1.4.
Lack of recognizing, respecting and promoting rights of the Indigenous Peoples	Under the regulatory surplus of the project activity, there are international rights such as UN Declaration on the Rights of Indigenous Peoples Rights and National rights such as the Land Act 1996 and Land Group Incorporation Act 1974 that are in place that ensures the Indigenous People have their rights and cultural heritage protected.

Any issues or problems observed can be brought up as a Grievance as explained more in section 2.1.4.

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken
No Risk Identified	This project is a voluntary project activity that involves the distribution of ICSs to individual households free of charge.

2.3.5 Benefit Sharing

Summary of the benefit sharing plan	Any household within the project boundary is eligible to be part of the project as long as they are effectively onboarded into the project activity first through the Household Registration Process and ICS Distribution Process. By completing these two steps, the end-user's household will receive an ICS free of charge and reap the direct and indirect benefits of using an ICS.
Benefit sharing during the monitoring period	The Household Registration Process and Distribution Process occurred between 23 September 2024 to 31 October 2024 during which 19,206 ICSs were distributed to unique households. Through the use of the ICS, end-users reaped the direct and indirect benefits of the ICS such as: - Reducing the amount of money and time spent buying or collecting firewood - Improvement of air-quality within the kitchen/cooking dwelling due to efficient burning of biomass - Reduction of forest degradation in the surrounding area due to the reduced need for biomass for cooking or heating For a full list of benefits, please refer to Section 1.12 of this document.

2.4 Ecosystem Health

Risk identified	Mitigation or preventative measure(s) taken during the monitoring period
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Impacts on biodiversity and ecosystems	No Risk Identified	The technology of the project involves the reduction of biomass for cooking or heating and will not impact the biodiversity and ecosystems in any way through soil degradation, erosion, and additional water consumption and stress.
Soil degradation and soil erosion	No Risk Identified	
Water consumption and stress	No Risk Identified	

2.4.1 Rare, Threatened, and Endangered species

As per the Project Design Document, the project is not located in or adjacent to habitats for rare, threatened, or endangered species.

Species or habitat	<i>Not applicable</i>
Areas needed for habitat connectivity	<i>Not applicable</i>

	Risks identified	Mitigation or preventative measure(s) taken
Habitats for rare, threatened, and endangered species	<i>Not applicable</i>	<i>Not applicable</i>
Areas for habitat connectivity	<i>Not applicable</i>	<i>Not applicable</i>

2.4.2 Introduction of species

The technology of the project involves the reduction of biomass for cooking or heating through ICS technology. It does not introduce any species of animal or plants into the project area, as such this section is not applicable.

Species introduced	Classification	Justification for use	Adverse effects and mitigation
<i>Not applicable</i>	<i>Not applicable</i>	<i>Not applicable</i>	<i>Not applicable</i>

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species
<i>Not applicable</i>	<i>Not applicable</i>

	Risks identified	Mitigation or preventative measure(s) taken
<i>Invasive species</i>	<i>Not applicable</i>	<i>Not applicable</i>

2.4.3 Ecosystem conversion

The technology of the project involves the reduction of biomass for cooking or heating through ICS technology. It does not conduct any ecosystem conversion within the project area; hence this section is not applicable.

	Risks identified	Mitigation or preventative measure(s) taken
<i>Ecosystem conversion</i>	<i>Not applicable</i>	<i>Not applicable</i>

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The first ICS was distributed on 23 September 2024, marking this date as the start date and commissioning of the project activity. A total of 19,206 ICSs was distributed during the monitoring period. During this period, there has been no reported exchanges or breakdowns of ICSs which may have impacted GHG emission reduction calculations.

The project proponent regularly visits end-users throughout the project's duration to gather feedback and assess the condition of the cookstoves. These visits offer opportunities to address any questions or concerns about the project, promote the continued use of the cookstoves, and provide training on their maintenance. However, there are no plans currently to replace any damaged devices and they will be conservatively removed from the GHG emissions reduction calculations.

3.2 Deviations

3.2.1 Methodology Deviations

The project activity is applying a one-off conservative deviation related to the use of the BC_p parameter. The BC_p was first measured in November 2022 and is valid for 2 years as per Section 9.2 of VM0050, v1.0, till November 2024, which is part-way through the current monitoring period. Values from the recent KPT measurement campaign concluded in July 2025 are to be used from December 2024 onwards for calculating ERs.

However, for the ease of efficiency in calculating the ERs from this monitoring period, the higher (more conservative) value of BC_p is uniformly applied throughout the monitoring period for each respective batch.

This deviation results in lower ERs, than if the BC_p values were separately applied for the monitoring period and is therefore more conservative and does not impact the applicability of the methodology, the project additionality, or the appropriateness of the baseline scenario.

3.2.2 Project Description Deviations

The project activity is applying a deviation for project description, to update the methodology from VMR0006 v1.2 to VM0050 v1.0. An updated project description document is prepared and made available for review along with this monitoring report. As per the VCS Standard v4.7 Section 3.21, projects are permitted to deviate from the validated project description in certain cases in order to accommodate changing circumstances post-validation.

The following requirements as shown in Table 5 for methodology deviation as per the VCS Standard have been applied in this project description.

Table 5: Project Description Deviations

S/N	Deviation Criteria	Project Applicability
1	Project proponents may apply project description deviations for the purpose of switching to a different methodology, where permitted. Where a project switches to a new methodology or methodology version, the project description shall be updated accordingly.	The project activity was initially registered under the VMR0006 v1.2 methodology and is updating to VM0050 v1.0. The project description has been updated to reflect the application of the latest applied methodology.
2	Projects cannot claim additional GHG emission reductions or carbon dioxide removals in a previously verified monitoring period resulting from a project description deviation.	The project activity has not begun any monitoring procedures for verification. During this methodology update, the monitoring plan will be updated to be applicable with VM0050 v1.0.
3	The deviation shall be assessed by a validation/verification body and the process, findings and conclusions shall be reported in the verification report. The assessment shall determine whether the deviation is appropriately described and justified, and whether the project remains in conformance with the VCS Program rules. The	The deviation will be assessed by a validation/verification body and the process, findings and conclusions shall be reported in the verification report. The assessment shall determine

	deviation shall also be reported on in all subsequent verification reports. Where the project description is updated, the updates shall be validated.	whether the deviation is appropriately described and justified, and whether the project remains in conformance with the VCS Program rules.
4	Project description deviations are not considered to be precedent setting.	It is confirmed that the project description deviations are not considered to be precedent setting.
5	The validation/verification body assessing the project description deviation shall be accredited for the validation, recognizing that assessment of project description deviations is a validation activity, as further set out in the VCS Program Guide.	The validation/verification body assessing the project description deviation is SustainCERT SA, an accredited validator for validation and recognizing that assessment of project description deviations is a validation activity, as further set out in the VCS Program Guide.

3.3 Grouped Projects

This monitoring report encompasses the 1st Project Instance (PI) of this Grouped Project Activity (GPA) and can be included into the GPA according to the eligibility criteria set out by the GPA in Table 6 and Table 7 below.

Table 6: Eligibility Criteria for Inclusion of New Project Instances

S/N	Criterion	Applicability	Evidence
1	Meets the applicability conditions set out in the methodology applied to the project.	New PIs will meet the applicability conditions set out in Section 4 of the methodology	- Project Design Document.
2	Use the technologies or measures specified in the project description.	Only ICSs that conform with the grouped project description are to be distributed in the project. The ICS will be chosen to	- ICS manufacturer's technology description.

		deliver a level of service at least equivalent to the baseline appliance.	
3	Applies the technologies or measures in the same manner as specified in the project description.	ICSs distributed in the Pls will adhere to the grouped project description.	- ICS manufacturer's Technical Specifications
4	Are subjected to the baseline scenario determined in the project description for the specified project activity and geographic area.	New Pls will be installed only in regions within the geographic borders of PNG subject to the same baseline scenario determined in Section 3.4 in this Project Description.	- Global Position System (GPS) coordinates captured from each end user will demonstrate the location of the project. - Virtual geographical log of ICS in monitoring database.
5	Have characteristics with respect to additionality that are consistent with the initial instances for the specified project activity and geographic area. For example, the new project activity instances have financial, technical and/or other parameters (such as the size/scale of the instances) consistent with the initial instances, or face the same investment, technological and/or other barriers as the initial instances.	All new Pls will use the activity method for demonstration of additionality. Step 1: Regulatory Surplus There is no government mandated programme or policy in host country of this project ensuring the distribution of new project activity instances. Step 2: Positive List The inclusion of new project activity instances will comply with the positive list as it satisfies criterion 1 where it meets all the	- Analysis on PNG government programs or policies for ICS conducted in Section 3.5. - The End User Agreement will confirm that the TEM “installs or distributes ICSs at zero cost to the end-user and has no other source of revenue other than the sale of GHG credits”.

		applicability conditions of the methodology.	
6	Occur within one of the designated geographic areas specified in the project description.	New PIs will only occur within the boundary of PNG.	- GPS Coordinates of ICS end user.
7	Comply with at least one complete set of eligibility criteria for the inclusion of new project activity instances. Partial compliance with multiple sets of eligibility criteria is insufficient.	New PIs will comply with at least one complete set of eligibility criteria for the inclusion into the GPA.	- Project Design Document.
8	Be included in the monitoring report with sufficient technical, financial, geographic, and other relevant information to demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body.	New PIs will demonstrate compliance with the applicable set of eligibility criteria and enable sampling by the validation/verification body through the inclusion of sufficient technical, financial, geographic, and other relevant information within the monitoring report.	- Sampling Plan - Monitoring Report
9	Be validated at the time of verification against the applicable set of eligibility criteria.	New PIs will be validated at the time of verification against the applicable set of eligibility criteria.	- Monitoring Report
10	Have evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance (i.e., the date upon which the project activity instance began reducing or removing GHG emissions).	New PIs will provide evidence of project ownership, in respect of each project activity instance, held by the project proponent from the respective start date of each project activity instance.	- The End User Agreement

11	Have a start date that is the same as or later than the grouped project start date.	The start date of new PIs will be after 23/09/2024, after the start date of the GPA.	- ICS Distribution Ledger
12	Be eligible for crediting from the start date of the instance through to the end of the project crediting period (only). Note that where a new project activity instance starts in a previous verification period, no credit may be sought for GHG emission reductions or removals generated during a previous verification period (as set out in Section 3.4.4) and new instances are eligible for crediting from the start of the next verification period.	New PIs will be eligible for crediting from the start date of the instance through to the end of the project crediting period (only). For example, when a PI starts with a fixed crediting period of 10 years (fixed), the project device/stove must be proven at inclusion that it will be able to sustain through the crediting period. This may be proven through the Project Device Lifespan Specification document.	- Monitoring Report - Project Device Lifespan Specification
13	Only eligible for crediting from the start of the verification period in which they were added to the grouped project.	Crediting Periods of New PIs will only start after the date that the project has been included into the grouped project.	- Monitoring Report
14	Not be or have been enrolled in another VCS project.	New PIs must declare that it has not or have been enrolled in another VCS project.	- Project Letter of Undertaking
15	Adhere to the clustering and capacity limit requirements for multiple project activity instances set out in 3.6.8 - 3.6.9 of the VCS Standard.	New PIs will adhere to the clustering and capacity limit requirements set out in Section 3.6.8 – 3.6.9 whereby the Project participant (PP) shall include all project activity	- GPS Coordinates of ICS end user.

		instances within ten kilometers of another instance of the same project activity and with the same project proponent and where a capacity limit applies to a project activity included in the project, no project activity instance shall exceed such limit.	
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Table 7:VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves v1.0

Methodology ID	Applicability condition	Justification of compliance
VM0050	The project activity corresponds to: a) Replacement of non-renewable biomass (e.g., firewood, charcoal)-fired cookstoves with any of the following: i) More efficient project devices that use the same fuel as in the baseline; ii) Efficient project devices fired by renewable biomass or bioethanol; iii) Efficient project devices fired by liquefied petroleum gas (LPG); or iv) Electric-powered project devices. b) Replacement of solid or liquid fossil fuel (e.g., coal, kerosene)-fired cookstoves with any of the following:	Applicable The project activity involves the replacement of non-renewable biomass with more efficient project devices that use the same fuel as in the baseline (firewood), therefore this criterion is met.

	i) Efficient project devices fired by renewable biomass or bioethanol; ii) Efficient project devices fired by LPG; or iii) Electric-powered project devices.	
VM0050	Project devices are used in households, communities, institutions, or small or medium enterprises (SMEs), collectively referred to in this methodology as the “target population.”	Applicable The project activity will be used in an end-user’s household for heating and cooking purposes. Therefore, this criterion is met.
VM0050	Where renewable biomass is used, it is exclusively renewable and qualifies as one of the following: a) A by-product, residue, or waste stream from agriculture, forestry, and related industries; or b) Originating from dedicated plantations that comply with all relevant applicability conditions in the most recent version of CDM TOOL16.	Does not Apply The project activity does not use renewable biomass therefore this criterion does not apply.
VM0050	Where biomass residues are used, they would have been left to decay or burned without energy recovery before implementation of the project activity, and their use does not involve a decrease in carbon pools – in particular of dead wood, litter, or soil organic carbon – on the land areas from which the biomass residues originate.	Does not Apply The project activity does not use biomass residues therefore this criterion does not apply.
VM0050	Where biomass residues from a production process are used, project implementation does not result in an increase in the processing capacity of raw	Does not Apply The project activity does not use biomass residues

	input or any other substantial changes (e.g., product change) in this process.	therefore this criterion does not apply.
VM0050	Where more than one type of renewable biomass is used, each of the biomass types used complies with the applicability conditions.	Does not Apply The project activity does not use renewable biomass therefore this criterion does not apply.
VM0050	Where project activities introduce renewable biomass such as charcoal, it is renewable charcoal produced by efficient charcoal production processes (e.g., retort sedentary kilns, improved sedentary kilns, Casamance kilns). Methane produced during the charcoaling process is captured and destroyed or combusted for energy purposes.	Does not Apply The project activity does not introduce renewable biomass such as charcoal into the project boundary therefore this criterion does not apply.
VM0050	Project devices using renewable biomass (fuel-switch) or non-renewable biomass (improved efficiency) are single-pot, multi-pot portable, or in-situ cookstoves with an initial thermal efficiency of at least 25%.	Applicable <i>The project activity uses non-renewable biomass single-pot portable cookstoves that has an initial thermal efficiency of 52%, therefore this criterion is met.</i>
VM0050	Project devices using LPG or bioethanol are single-pot, multi-pot portable, or in-situ cookstoves with an initial thermal efficiency of at least 30%.	Does not Apply The project activity uses ICSs that utilize firewood as a fuel and does not involve LPG or bioethanol therefore this criterion does not apply.
VM0050	Electric project devices meet the maximum risk factor score of 15 on the Cookstove Durability Protocol and have the following minimum thermal efficiency: a) Hot plates and electric hobs: 40% b) Induction stoves and other electric stoves: 70%	Does not Apply The project activity uses ICSs that utilize firewood as a fuel for cooking and heating purposes, therefore this criterion does not apply.

VM0050	Project devices using LPG comply with all of the following conditions: a) The baseline fuel either includes non-renewable biomass or is a more carbon-intensive fossil fuel. b) The project has a provision for metering LPG supplied to each consumer at the LPG filling station, in order to determine household LPG consumption; and c) The project does not seek to issue any carbon credits for periods after 31 December 2045.	Does not Apply The project activity uses ICSs that utilize firewood as a fuel for cooking and heating purposes, therefore this criterion does not apply.
VM0050	Electric project devices use the following electricity sources: a) Decentralized renewable energy systems: Decentralized energy systems using fossil fuels are not eligible, except for backup generators that supply less than 1% of the annual electricity of the decentralized renewable energy system. b) Self-generated renewable electricity (with a maximum of 20% electricity from non-renewable sources for backup); or c) National or regional electricity grid.	Does not Apply The project activity uses ICSs that utilize firewood as a fuel for cooking and heating purposes, therefore this criterion does not apply.
VM0050	The project proponent designs incentive mechanisms to reduce the use of inefficient baseline devices and practices that can be replaced by the project devices and describes these mechanisms in the project description.	Applicable The project proponent employs the following incentive methods to reduce the use of inefficient baseline devices and practices:

		<u>Household Education and Encouragement:</u> The project proponent, through cookstove ambassadors, routinely visits end-users throughout the project lifetime to collect feedback, maintain the project cookstoves, and address questions or queries regarding the project activity to encourage the continued use of the project cookstoves. <u>Behavioural Change Campaigns:</u> Awareness campaigns are also held by field assistants and cookstove ambassadors in the form of community meetings featuring cooking demonstrations, peer testimonials within villages to promote and educate households on the health, environmental, and economic benefits of transitioning to improved cookstoves, thereby motivating voluntary behaviour change. Additionally, the project proponent works with the Western Pacific University (WPU) of Papua New Guinea to develop and train locals on project management and carbon knowledge. Graduates of this course would be able to work with the project proponent formally during the project lifetime.
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		<u>Monitoring and Enforcement:</u> During the project lifetime, visits done for household education and encouragement will also monitor the usage rate of the ICS. This effectiveness of the abovementioned incentive mechanisms will be monitored as a decrease in stove stacking factor. <u>Additional Incentive Engagement:</u> From the results of the abovementioned monitoring, villages showing the most progress will be rewarded with celebratory ‘Cook-Off’ Events featuring demonstrations, knowledge-sharing, and success stories. Testimonials will be gathered for short-form videos to boost awareness via social media, while non-adopters will be consulted to identify and address adoption barriers for future training improvements. Hence, this criterion is met.
VM0050	Where a project device has ended its technical life, the project proponent either replaces it with a comparable or better project device or retrofits its essential components to continue meeting the minimum service level requirements (i.e., thermal energy generation), otherwise no further emission reductions may be claimed for the project device.	Applicable Where a project device has ended its technical life, the project proponent does not replace the ICS. Instead, no further emission reductions will be claimed for the project device, therefore this criterion is met.

VM0050	Project proponents implement a method for the distribution and identification of project devices that avoids double counting of emission reductions by other mitigation actions and includes unique serial numbers on the stove itself at the time of distribution/sale (e.g., program logo, alpha/numeric ID, and end-user location, such as geographic coordinates, complete address information).	Applicable In order to avoid double counting emission reductions through the method of distribution, each ICS will carry a Unique Serial Number (USN), and the distribution of each device is logged into an electronic database which includes unique information for each recipient (i.e., Unique Serial Number, geographic coordinates, etc.) upon receipt of the device. The electronic database combined with internal checks done by the PP will ensure that no UINs are double counted. During the monitoring period, survey samples will be only taken from the USNs that are in the electronic database of end-users eliminating the risk of double counting from any other projects within the same geographical location.
VM0050	The project complies with any national, sub-national, or local regulations or guidance for the installation, commercialization, distribution, and use of ICSs and/or fuel supply and use for the target population. National, regional, and local regulatory frameworks for the provision of the type of thermal energy services provided by the project activity must be documented.	Applicable The project activity complies with any national, sub-national, or local regulations or guidance for the installation, commercialization, distribution, and use of ICS and/or fuel supply and use for the target population. At every stage of project inception, the project proponent has been in touch with all levels of relevant authorities and have gained

		approval, therefore this criterion is met.
VM0050	Where project activities reduce emissions from non-renewable biomass, including firewood and charcoal, the risk of double counting is assessed on a national basis by evaluating at validation and crediting period renewal whether there are REDD+ projects or jurisdictional REDD+ programs whose project boundary overlaps with the expected fuel source area of the project. The project proponent must report on the findings of this assessment for informational purposes in the project description.	Applicable Currently, there are no REDD+ projects or jurisdictional REDD+ programs with the boundary of PNG Highlands, therefor this criterion is applicable.

3.4 Baseline Reassessment

Did the project undergo baseline reassessment during the monitoring period?

☐ Yes ☒ No

The project activity was formally registered on 14/06/2024 under the methodology VMR0006: Energy Efficiency and Fuel Switch Measures in Thermal Applications, v1.2. and underwent a methodology update to VM0050: Energy Efficiency and Fuel-Switch Measures in Cookstoves, v1.0 as per Verra Guidelines²³.

In accordance with the *Procedure to Change Methodology through a Project Description Deviation v4.0*, the project is not required to undergo a baseline reassessment, as its technology does not fall under any of the categories listed in Section 2.1.3, item 1, footnote 3. Instead, as outlined in the procedure for projects not subject to baseline reassessment, TEM has applied the new methodology to determine the most plausible baseline scenario and to calculate baseline emissions for the remainder of the crediting period, in line with the revised baseline scenario requirements. The selection and justification of the baseline scenario is covered in Section 3.4 of the PD.

Established Controlled Households

As per VM0050 v1.0 Section 6.2 and Correction 1, follow-up baseline surveys must be conducted every 5-years in controlled households and do not participate in the project. These control households must be established prior to validation and must be shown to be statistically

²³ VM0050 Energy Efficiency and Fuel-Switch Measures in Cookstoves, v1.0, <https://verra.org/methodologies/vm0050-energy-efficiency-and-fuel-switch-measures-in-cookstoves-v1-0/>

equivalent to the pre-project conditions of project households regarding baseline fuel type(s) and percentage of use by the target population, source(s) of each baseline fuel, and baseline technologies for cooking.

The control houses are identified, however not registered on a project database, due to practical on-the-ground operational concerns that can cause risks to the longevity of the project. House registrations trigger an anticipation of cookstoves to be distributed imminently, while by design we are not distributing cookstoves in these households. Over time, this expectation can escalate into frustrations and can affect the safety and harmony of the field teams working in the area and other neighbouring houses where project devices were distributed.

Therefore, it is not practical to register all the households on a project database, and instead an entire ward, which is the smallest administrative district in PNG, is chosen as a control. The control households are identified to be in the Koalilombo ward²⁴ of the Southern Highlands Province within the project boundary. Based on PNG Census 2011²⁵, this ward has an estimated 341 households and a population of approximately 1,992 people. This area has the same baseline and is within the project boundary, and is in close proximity to the cookstove distribution areas, and therefore a suitable control area.

No ICSs were distributed to households within this ward during this monitoring period. A KML file demarcating the geographical extend of Koalilombo ward is provided to the VVB to validate that no cookstoves are distributed in the region during the monitoring period.

²⁴ GPS Coordinates: 6° 25' 56.7" S 143° 54' 11.2" E, <https://maps.app.goo.gl/6J9omZzdJf47jdHL8>

²⁵ National Statistics Office, Papua New Guinea National Population & Housing Census 2011, 'Count me in' (2011)

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	EF _{b,i,CO2} EF _{p,j,CO2}
Data unit	tCO ₂ /TJ
Description	CO ₂ emission factor for fuel used by baseline device type i in the baseline scenario. CO ₂ emission factor for fuel used by project device type j in the project Scenario.
Source of data	2019 IPCC Guidelines for National Greenhouse Gas Inventories, Chapter 2, Stationary Combustion
Value applied	112
Justification of choice of data or description of measurement methods and procedures applied	Methodology Default
Purpose of data	Calculation of baseline and project emissions
Comments	N/A

Data / Parameter	EF _{b,i,nonCO2} EF _{p,j,nonCO2}
Data unit	tCO _{2e} /TJ
Description	Non-CO ₂ emission factor for fuel used by baseline device type i in the baseline scenario. Non-CO ₂ emission factor for fuel used by project device type j in the project scenario.
Source of data	2019 IPCC Guidelines for National Greenhouse Gas Inventories, Chapter 2, Stationary Combustion

Value applied	9.46
Justification of choice of data or description of measurement methods and procedures applied	Methodology Default
Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	NCV _{b,i} NCV _{p,j}
Data unit	TJ/tonne
Description	Net calorific value of baseline fuel used by baseline device type i Net calorific value of project fuel used by project device type j
Source of data	2019 IPCC Guidelines for National Greenhouse Gas Inventories, Chapter 1, Introduction
Value applied	0.0156
Justification of choice of data or description of measurement methods and procedures applied	2019 IPCC default for wood fuel, 0.0156 TJ/tonne, based on the gross weight of the wood that is 'air-dried' may be used if fuel used in project device is woody biomass/renewable biomass.
Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	BC _{ex-ante,b,i}
Data unit	Tonnes
Description	Ex-ante annual average quantity of fuel used per baseline device type i
Source of data	TEM 2025_Cookstoves Highlands region report_v3.pdf and audited by SPFEC Letter to TEM-Review of Cookstoves Highlands region report_17-01-2025.pdf
Value applied	5.24

Justification of choice of data or description of measurement methods and procedures applied	The quantity of fuel that would be used in the baseline scenario was determined through Option 1: Measurement Campaign conducted by a third-party consultant team contracted by TEM along with the baseline study survey concluded in November 2022. A total of 220 houses were selected for the baseline study with 110 houses each from the Southern Highlands Province and Western Highlands Province. The selected houses were provided with an individual code from 001 to 220. Of the total number of houses, 160 were used for control cooking tests (CCT) comparing firewood consumption using ICSs (ICS, n=80) with traditional open fire cooking (OFC, n=80). At the end of the campaign, sampling challenges enabled the collection of firewood consumption data on 131 households in both provinces, of which 60 houses used ICSs and the remaining 71 houses used traditional OFC. The results of the measurement campaign yielded a value of 5.97 tonnes as the firewood usage per year on a baseline stove (OFC) (file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf). As the average daily fuel consumption per adult equivalent parameter from the measurement campaign was obtained from 71 samples and achieved a precision value 12.36%, the lower bound value of 5.24 tonnes/year is used after applying a conservativeness deduction as per Section 4 of CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities, version 9.0.
Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	$\eta_{old,avg}$
Data unit	Fraction
Description	Weighted average efficiency of baseline devices that are replaced by project devices
Source of data	Methodology Default

	As per the applied methodology, for three-stone fire using firewood or a cookstove with no improved combustion air supply or flue gas ventilation, default value of 15%.
Value applied	0.15
Justification of choice of data or description of measurement methods and procedures applied	Methodology Default, CDM TOOL33_v2 During the baseline study TEM found that the prevailing pre-project device was a traditional three stone fire with no improved combustion air supply or flue gas ventilation, and thus, no efficiency tests were carried out (see file titled; TEM 2025_Cookstoves Highlands region report_v3.pdf).
Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / Parameter	Hh _i Hh _{j,k}
Data unit	Equivalent standard male adults
Description	Average household size of the target population using device type i Average household size of the target population using device type j from batch k
Source of data	TEM 2025_Cookstoves Highlands region report_v3.pdf and audited by SPFEC Letter to TEM-Review of Cookstoves Highlands region report_17-01-2025.pdf
Value applied	4.85
Justification of choice of data or description of measurement methods and procedures applied	The baseline report conducted is in line with the CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities v9.0 and has achieved a confidence and precision of at least 90/10 for the target parameter of average household size.
Purpose of data	Estimation of average energy consumption when applying <u>Option 1: Measurement campaign</u> Cross-checking energy and fuel consumption values
Comments	N/A

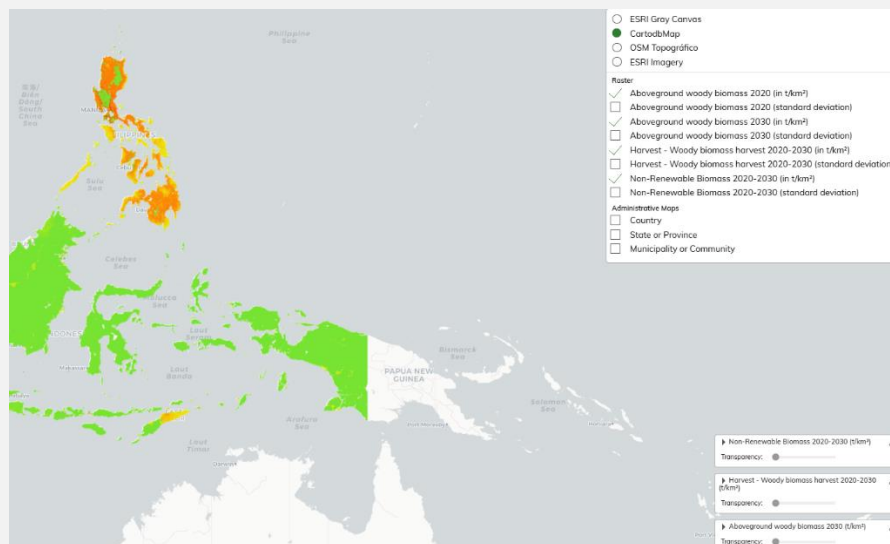
Data / Parameter	fNRB _{b,y}																									
Data unit	Fraction																									
Description	Fraction of woody biomass that can be established as non-renewable biomass.																									
Source of data	TEM (2023). f _{NRB} calculation_Final (see letter titled: GeoNarem Letter for fNRB_Jan 23.pdf) audited by the Head of Forestry of PNG University of Technology (file titled: SPFEC Letter to TEM-Review of F _n RB Calculations.pdf).																									
Value applied	0.60																									
Justification of choice of data or description of measurement methods and procedures applied	Calculated as per the CDM Methodological Tool 30 Version 4: Calculation of the fraction of non-renewable biomass. fNRB = NRB/(NRB+RB) = 81.24% Applying a discount of 26% = 60.12% <table border="1"> <thead> <tr> <th>Parameter</th><th>Value</th><th>Source</th></tr> </thead> <tbody> <tr> <td>NRB = H - RB</td><td>25,844,547</td><td>Calculated value</td></tr> <tr> <td>RB = MAI x (F - P)</td><td>5,967,605</td><td>Calculated value</td></tr> <tr> <td>MAI (Mean Annual Increment, t/ha/yr)</td><td>0.69</td><td>PNG's Second Biennial Update Report to UNFCCC 2022, Table 2-8, pg 25</td></tr> <tr> <td>F (Extent of forest, ha)</td><td>35,963,272</td><td>Papua New Guinea Forest Authority (2019). Forest and Land Use Change in PNG 2000-2015, Table 7-1, p71</td></tr> <tr> <td>P (Extent of non-accessible area, ha)</td><td>27,296,091</td><td>Papua New Guinea Forest Authority (2019). Forest and Land use change in PNG 2000-2015, p35.</td></tr> <tr> <td>H = HW x N + CE + NE</td><td>31,812,152</td><td>Calculated value</td></tr> <tr> <td>HW</td><td>6.8</td><td></td></tr> </tbody> </table>		Parameter	Value	Source	NRB = H - RB	25,844,547	Calculated value	RB = MAI x (F - P)	5,967,605	Calculated value	MAI (Mean Annual Increment, t/ha/yr)	0.69	PNG's Second Biennial Update Report to UNFCCC 2022, Table 2-8, pg 25	F (Extent of forest, ha)	35,963,272	Papua New Guinea Forest Authority (2019). Forest and Land Use Change in PNG 2000-2015, Table 7-1, p71	P (Extent of non-accessible area, ha)	27,296,091	Papua New Guinea Forest Authority (2019). Forest and Land use change in PNG 2000-2015, p35.	H = HW x N + CE + NE	31,812,152	Calculated value	HW	6.8	
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HW	6.8																									

	(Average consumption of wood fuel per household, t)		Lit. reviews from Page et. al (2016), ACIAR (2013), Murphy et. al (2009), and TEM survey in project area (2023, unpublished)
	N (Number of households in surveyed area)	197,869	
	CE + NE (Commercial woody biomass consumption for energy and non-energy applications respectively, t)	30,466,643	Data gathered from Papua New Guinea Forest Authority (2019). Forest and Land use change in PNG, 2000 -2015 and PNG's Second Biennial Update Report to UNFCCC 2022
Purpose of data	Calculation of emission reductions.		
Comments	The MoFuSS national and sub-national estimations for fNRB, recently adopted by UNFCCC (Annex 3, MP96, March 2025), does not have country specific values for key parameters including above ground biomass, harvest rate and non-renewable biomass in Papua New Guinea in the current iteration as seen from the screenshot from the MoFuSS database below. In addition, the firewood demand is modelled²⁶ for the years 2020 to 2030 (and beyond) using data from <u>Stoner et.al (2021)</u> which models macro-trends in firewood usage globally from a WHO dataset of air pollution data²⁷. This single extrapolated value for the entire country of Papua New Guinea does not take into account the substantial regional variation in fuelwood usage. The Highlands region hosts 40% of the national population in densely populated areas that are characterized by the lowest average temperatures leading to the highest consumption of firewood in PNG (Hanson et al., 2001; ACIAR, 2013; Nuberg, 2015). This		

²⁶ <https://www.mofuss.unam.mx/tools/apps/wds/>

²⁷ <https://www.who.int/data/gho/data/themes/air-pollution/household-air-pollution>

regional variation is explored in detail in Section 3.9 of the baseline study (file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf)



TOOL 30 was ultimately chosen for estimating the fNRB due to the lack of accurate published datasets for PNG, and specifically the Highlands region. All the data used in the CDM Tool 30 calculation (file titled: TEM 2023_fNRB calculation_Final) were obtained from PNG national statistics reported by the PNGFA (2019) including the forest regrowth rates from Biennial Updates submitted to UNFCCC in 2022. These in country specific datasets represent the most up-to-date literature providing more accuracy in the fNRB calculations than the proposed pixel-based models used in MoFuSS. This has been independently verified by an in-county forestry expert (file titled: SPFEC Letter to TEM-Review of FNRB Calculations.pdf).

Data / Parameter	Life Span
Data unit	Number of years
Description	The operating lifetime of the project device. The life span should be reported if the methodology equation 5 is adopted to determine the ICS's efficiency
Source of data	Manufacturer's specification
Value applied	10

Justification of choice of data or description of measurement methods and procedures applied	This parameter shall be determined ex-ante
Purpose of data	Calculation of emission reductions.
Comments	N/A

4.2 Data and Parameters Monitored

Data / Parameter	$N_{j,1,y}$
Data unit	Number
Description	Number of commissioned project devices of type j from batch 1 in year 1
Source of data	Distribution Ledger
Description of measurement methods and procedures to be applied	Distribution Ledger will be validated each time a new batch of ICSs are distributed as part of the project instance. Example: If the project instance takes 3 years to fully distribute all ICSs, monitoring for this parameter will only be conducted for the first 3 years.
Frequency of monitoring/recording	Yearly, when there are new commissioned ICSs
Value applied	4,649
Monitoring equipment	N/A
QA/QC procedures to be applied	The project implementor shall maintain a distribution record to calculate this parameter.
Purpose of data	Calculation of baseline and project emissions
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value.
Comments	N/A
Data / Parameter	$N_{j,2,y}$

Data unit	Number
Description	Number of commissioned project devices of type j from batch 2 in year 1
Source of data	Distribution Ledger
Description of measurement methods and procedures to be applied	Distribution Ledger will be validated each time a new batch of ICSs are distributed as part of the project instance. Example: If the project instance takes 3 years to fully distribute all ICSs, monitoring for this parameter will only be conducted for the first 3 years.
Frequency of monitoring/recording	Yearly, when there are new commissioned ICSs
Value applied	14,557
Monitoring equipment	N/A
QA/QC procedures to be applied	The project implementor shall maintain a distribution record to calculate this parameter.
Purpose of data	Calculation of baseline and project emissions
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value.
Comments	N/A

Data / Parameter	Date of commissioning of batch 1
Data unit	Date
Description	To establish the date of commissioning, the Project Participant may opt to group the devices in “batches” and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch
Source of data	Internal Log of Project device registrations.
Description of measurement methods and procedures to be applied	N/A

Frequency of monitoring/recording	Fixed and recorded at the time of commissioning/distribution of the last project device in the batch
Value applied	30 September 2024
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of emission reductions.
Calculation method	N/A
Comments	To be reported in the monitoring report

Data / Parameter	Date of commissioning of batch 2
Data unit	Date
Description	To establish the date of commissioning, the Project Participant may opt to group the devices in “batches” and the latest date of commissioning of a device within the batch shall be used as the date of commissioning for the entire batch
Source of data	Internal Log of Project device registrations.
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	Fixed and recorded at the time of commissioning/distribution of the last project device in the batch
Value applied	31 October 2024
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of emission reductions.
Calculation method	N/A
Comments	To be reported in the monitoring report

Data / Parameter	Date of commissioning of project device i
Data unit	Date
Description	Actual date of commissioning of the project device
Source of data	Internal Log of Project device registrations.
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	Fixed and recorded at the time of commissioning/distribution
Value applied	23 September 2024
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of emission reductions.
Calculation method	N/A
Comments	To be reported in the monitoring report

Data / Parameter	$n_{j,1,y}$
Data unit	Fraction
Description	Proportion of commissioned project devices of type j from batch 1 that are still being used regularly in year y
Source of data	Monitoring survey
Description of measurement methods and procedures to be applied	<u>Option 2 (surveys):</u> The total number of devices in operation compared to the total number distributed (according to the monitoring database) will be surveyed. Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 90/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and programme of activities.

Frequency of monitoring/recording	Annually
Value applied	0.84
Monitoring equipment	Monitoring Survey
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of baseline and project emissions
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value.
Comments	Monitoring Survey results for this monitoring period reported 100% usage rate, however, as per Clarification 2 of VM0050 v1.0 Correction and Clarifications ²⁸ , the usage rate has been capped at 90%.

Data / Parameter	$n_{j,2,y}$
Data unit	Fraction
Description	Proportion of commissioned project devices of type j from batch 2 that are still being used regularly in year y
Source of data	Monitoring survey
Description of measurement methods and procedures to be applied	<u>Option 2 (surveys):</u> The total number of devices in operation compared to the total number distributed (according to the monitoring database) will be surveyed. Measured directly or based on a representative sample. Sampling standard shall be used for determining the sample size to achieve 90/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and programme of activities.
Frequency of monitoring/recording	Annually
Value applied	0.9

²⁸ https://verra.org/wp-content/uploads/2025/02/CC_VM0050_v1.0_Feb2025.pdf

Monitoring equipment	Monitoring Survey
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of baseline and project emissions
Calculation method	Proportion of operational stoves obtained from the survey is multiplied by the total commissioned stoves to arrive at this value.
Comments	Monitoring Survey results for this monitoring period reported 100% usage rate, however, as per Clarification 2 of VM0050 v1.0 Correction and Clarifications ²⁹ , the usage rate has been capped at 90%.

Data / Parameter	BC _{b,i,y}
Data unit	Tonnes
Description	Fuel used per baseline device type i during year y
Source of data	Option 1: A measurement campaign following the Kitchen Performance Test Protocol in control households that do not participate in the project, established prior to validation as statistically equivalent to the pre-project conditions of project households regarding baseline fuel consumption
Description of measurement methods and procedures to be applied	The quantity of fuel that would be used in the baseline scenario was determined through Option 1: Measurement Campaign conducted by a third-party consultant team contracted by TEM along with the baseline study survey concluded in November 2022. A total of 220 houses were selected for the baseline study with 110 houses each from the Southern Highlands Province and Western Highlands Province. The selected houses were provided with an individual code from 001 to 220. Of the total number of houses, 160 were used for control cooking tests (CCT) comparing firewood consumption using ICSs (ICS, n=80) with traditional open fire cooking (OFC, n=80). At the end of the campaign, sampling challenges enabled the collection of firewood consumption data on 131 households in both provinces, of which 60 houses used ICSs and the remaining 71 houses used traditional OFC. The results of the measurement campaign

²⁹ https://verra.org/wp-content/uploads/2025/02/CC_VM0050_v1.0_Feb2025.pdf

	yielded a value of 5.97 tonnes as the firewood usage per year on a baseline stove (OFC) (file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf). As the average daily fuel consumption per adult equivalent parameter from the measurement campaign was obtained from 71 samples and achieved a precision value 12.36%, the lower bound value of 5.24 tonnes/year is used after applying a conservativeness deduction as per Section 4 of CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities, version 9.0.
Frequency of monitoring/recording	Every five years
Value applied	5.24
Monitoring equipment	Monitoring Surveys
QA/QC procedures to be applied	As the average daily fuel consumption per adult equivalent parameter from the measurement campaign was obtained from 71 samples and achieved a precision value 12.36%, the lower bound value of 5.24 tonnes/year is used after applying a conservativeness deduction as per Section 4 of CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities, version 9.0.
Purpose of data	Calculation of baseline emissions
Calculation method	N/A
Comments	N/A

Data / Parameter	$\eta_{\text{new,avg,y}}$
Data unit	Fraction
Description	Weighted average efficiency of project devices in year y
Source of data	Monitoring Surveys, Manufacturer specifications on efficiency based on water boiling test (WBT).
Description of measurement methods and procedures to be applied	As per VM0050 v 1.0, the efficiency of project devices will be established via the following method: 1. Manufacturer-certified value that is determined via Water Boiling Test;

	The decrease in thermal efficiency of project devices from their respective batches due to aging must be accounted for during the monitoring period. As per the AMS-II.G., the loss in efficiency has to be monitored annually through the Standard Water Boiling Test campaign from a respective sample of each batch. The actual loss rate is then measured and used. This is not applicable for this monitoring period which is less than 1 year from the project start date, as all ICSs are newly distributed and efficiency would be assumed to be manufacturer-certified value.
Frequency of monitoring/recording	1) Recorded at the time of commissioning/distribution then; 2) Annually
Value applied	0.524
Monitoring equipment	Monitoring survey, WBTs
QA/QC procedures to be applied	The weighted average efficiency of project devices is first established via a Manufacture-certified value that is determined via a Water-Boiling test. Subsequently, thermal efficiency due to aging is accounted annually via Standard Water Boiling Test campaigns as per AMS-II.G. v13.1, where the loss in efficiency must be monitored annually through the Standard Water Boiling Test campaign from a respective sample of the first batch. The actual loss rate is then measured and used. The following simplified approach may be used, when the efficient ICSs are produced by a manufacturer with a recognized management system in place (e.g. ISO certification) to ensure that the individual equipment produced do not vary beyond the range of acceptance limits (e.g. characteristics such as materials, critical dimensions): 1. Conduct a sample test on three ICSs with three tests conducted for each ICS. The test can be carried out by project proponents by themselves or ICS manufacturers; If the standard deviation of the nine test results indicated above is very small and 90/10 precision requirement is met (in this case, the value of the t-distribution for 90 per cent confidence shall be used instead of Z value), the efficiency determined is acceptable, otherwise more sample tests would be required until 90/10 precision is met. This is not applicable for this monitoring period which is less than 1 year from the project start date, as all ICSs are newly distributed and efficiency would be assumed to be manufacturer-certified value.
Purpose of data	Calculation of project emissions

Calculation method	N/A
Comments	N/A

Data / Parameter	$BC_{p,j,1,y}$
Data unit	Tonnes
Description	Average quantity of fuel used by project device type j from batch 1 during year y
Source of data	Measurement campaign as mentioned in TEM 2025_Cookstoves Highlands region report_v3.pdf and audited by SPFEC Letter to TEM-Review of Cookstoves Highlands region report_17-01-2025.pdf
Description of measurement methods and procedures to be applied	$BC_{p,j,k,y}$ is estimated using <u>Option 1: Kitchen Performance Test</u> wherein measurement campaigns were concluded during baseline survey in November 2022 and more recently in July 2025. The project is applying a one-off deviation for the applicability of this parameter, see Section 3.2.1. As the parameter BC_p was first measured in November 2022 and is valid for 2 years as per Section 9.2 of VM0050, v1.0, till November 2024, which is part-way through the current monitoring period. Values from the recent KPT measurement campaign concluded in July 2025 are to be used from December 2024 onwards for calculating ERs. However, for the ease of efficiency in calculating the ERs from this monitoring period, the higher (more conservative) value of BC_p is uniformly applied throughout the monitoring period for each respective batch. This deviation results in lower ERs, than if the BC_p values were separately applied for the monitoring period and is therefore more conservative and does not impact the applicability of the methodology, the project additionality, or the appropriateness of the baseline scenario. Value from November 2022 KPT: 1.54 tonnes/year Value from July 2025 KPT for batch 1: 1.61 tonnes/year (more conservative, applied for the monitoring period)

Frequency of monitoring/recording	Annually
Value applied	1.61
Monitoring equipment	N/A
QA/QC procedures to be applied	For the November 2022 KPTs, the average daily fuel consumption per adult equivalent parameter from the measurement campaign was obtained from 60 samples and achieved a precision value 12.98%, the upper bound value of 1.54 tonnes/year is used after applying a conservativeness deduction as per Section 4 of CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities, version 9.0. For the recent KPTs concluded in July 2025, a mean value of 1.61 tonnes/year was obtained from a total of 50 samples with a precision value of 7.45%, and no conservativeness deduction is needed.
Purpose of data	Calculation of baseline emissions
Calculation method	N/A
Comments	N/A

Data / Parameter	$BC_{p,j,2,y}$
Data unit	Tonnes
Description	Average quantity of fuel used by project device type j from batch 2 during year y
Source of data	Measurement campaign as mentioned in TEM 2025_Cookstoves Highlands region report_v3.pdf and audited by SPFEC Letter to TEM-Review of Cookstoves Highlands region report_17-01-2025.pdf
Description of measurement methods and procedures to be applied	$BC_{p,j,k,y}$ is estimated using <u>Option 1: Kitchen Performance Test</u> wherein measurement campaigns were concluded during baseline survey in November 2022 and more recently in July 2025. The project is applying a one-off deviation for the applicability of this parameter, see Section 3.2.1. As the parameter BC_p was first measured in November 2022 and is valid for 2 years as per

	Section 9.2 of VM0050, v1.0, till November 2024, which is part-way through the current monitoring period. Values from the recent KPT measurement campaign concluded in July 2025 are to be used from December 2024 onwards for calculating ERs. However, for the ease of efficiency in calculating the ERs from this monitoring period, the higher (more conservative) value of BC_p is uniformly applied throughout the monitoring period for each respective batch. This deviation results in lower ERs, than if the BC_p values were separately applied for the monitoring period and is therefore more conservative and does not impact the applicability of the methodology, the project additionality, or the appropriateness of the baseline scenario. Value from November 2022 KPT: 1.54 tonnes/year (more conservative, applied for the monitoring period) Value from July 2025 KPT for batch 1: 1.49 tonnes/year
Frequency of monitoring/recording	Annually
Value applied	1.54
Monitoring equipment	N/A
QA/QC procedures to be applied	As the average daily fuel consumption per adult equivalent parameter from the measurement campaign was obtained from 60 samples and achieved a precision value 12.98%, the upper bound value of 1.54 tonnes/year is used after applying a conservativeness deduction as per Section 4 of CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities, version 9.0. For the recent KPTs concluded in July 2025, a mean value of 1.49 tonnes/year was obtained from a total of 42 samples with a precision value of 8.23%, and no conservativeness deduction is needed.
Purpose of data	Calculation of baseline emissions
Calculation method	N/A
Comments	N/A

4.3 Monitoring Plan

Sampling Plan

The sampling plan is intended to enable the discovery of unbiased and reliable estimates of monitored parameter values used in the GHG emission reduction calculations. The sampling plan design is based on the CDM Guideline: Sampling and surveys for CDM project activities and programmes of activities, version 4.0, and will be designed to monitor the parameters listed in the Section above, which are required for calculation of the actual GHG emission reduction achieved by the project activity using ex post sampling survey.

The share of operating ICSs and the continued use of pre-project devices will be determined based on sampling procedures as outlined below. The PP will be responsible for conducting the sampling surveys and maintaining a database with all operating ICSs.

As per the section 6 of the Guideline for Sampling and Surveys for CDM Project Activities and Programme of Activities, version 04, the sampling plan is the following:

1) Sampling Design

Due to the large number of ICSs envisioned to be distributed as part of the project activity, it is not economically feasible to monitor each individual ICS unit distributed. Therefore, representative sampling will be undertaken as part of a project instance-wide Sampling Plan (by grouping and sampling across project activities) that is designed in line with the requirements of the “Sampling and surveys for CDM project activities and programme of activities”, version 04.0.

1.1) Objective and Reliability Requirements

Sampling activities are defined at the Grouped Project-level and the sampling plan presented here will apply to assess parameter values in Project Instances to be included in the Grouped Project, either individually or in groups. These parameters are described more in Table 8.

Table 8: Monitored Parameters

Parameter and Type	Parameter and Type	Indicative Timing	Frequency of Monitoring Required by Methodology	Methods to be Applied
$\eta_{j,k,y}$ (Proportion of commissioned project devices)	Fraction	Monitoring will likely occur at least once every 12 months	Annually	Field staff visual inspection of the ICS end users, and questionnaires

of type j from batch k that are still being used regularly in year y)				
$BC_{b,i,y}$	Tonnes	Every five years	Every five years	Field staff visual inspection of the ICS end users, and questionnaires
$BC_{p,j,k,y}$	Tonnes	Biennially	Biennially	The more conservative Kitchen Performance Tests (KPT) conducted in 2022 and 2025. Please see deviation in Section 3.21.
$\eta_{new,avg,y}$	Fraction	- Recorded at the time of commissioning/distribution then; - Annually	Annually	- Efficiency of project devices is established via a manufacturer-certified value. - Decrease in thermal efficiency due to aging is monitored annually through the Standard Water Boiling Test campaign.

2.1) Target Population

- The target population for the proportion of ICSs still in operation ($n_{j,k,y}$) of this project are all households in the project database which are using fuel wood in ICSs distributed under the project for cooking.
- The target population for fuel used per baseline device ($BC_{b,i,y}$) is set in control households that do not participate in the project, established prior to validation as statistically equivalent to the pre project conditions of project households regarding baseline fuel consumption.
- The target population for average quantity of fuel consumed by the project device ($BC_{p,j,k,y}$) are all households in the project database which are using fuel wood in ICSs distributed under the project for cooking.

- The target population for the weighted average efficiency of project devices ($\eta_{\text{new,avg,y}}$) is established via a Manufacture-certified value determined via Water-Boiling Test. Subsequently, the decrease in thermal efficiency of project devices from their respective batches due to aging must be accounted for during the monitoring period. As per AMS-II.G., the loss in efficiency has to be monitored annually through the Standard Water Boiling Test campaign from a respective sample of each batch. The actual loss rate is then measured and used.

2.2) Sampling Frame

The projects are to be implemented in rural/semi-urban areas; thus, it is expected that the geographical locations do not have influence on the parameter of interest. Therefore, all above mentioned parameters can be assumed to be highly homogeneous for each ICS model regardless of how the end user group and distribution/installation location is defined.

1) Sampling frame for proportion of ICSs in operation ($n_{j,k,y}$)

The sample frame refers to all the information sources on the Database. There are two primary mechanisms for data collection: the Registration Process for newly distributed/installed ICSs and the Monitoring Survey (which includes a household questionnaire and visual inspection of ICSs) that will be used throughout the lifetime of the ICS. The detailed information collected from Registration Process is used to populate the ICS Database and the Monitoring Survey follows “Sampling and Surveys for CDM Project Activities and Programme of Activities”, version 04.

As explained below (on section “Sampling Method”), to take the different characteristics of different project instances Implemented and ICS models into consideration, Project instances shall be grouped together to create a Primary Sampling Unit, which is homogenous. As per EB 86 Annex 04, Appendix-2, paragraph 1, for the use of a single sampling plan covering a group of projects, provided the homogeneity of population can be demonstrated, or differences are considered in the sample size calculation, a 90/10 confidence/precision is applied for biennial sampling.

The first step is to identify the Primary Sampling Units. Primary sampling units are project instances, which have:

Identification of Primary Sampling Units:

Primary sampling units are project instances, which have:

- The same ICS models

This project with the same ICS model can therefore be grouped together and form a Primary Sampling Unit.

2.3) Sampling Method

For each survey, the sampling method for monitored parameters is determined by using a Simple Random Sample of the total population of distributed ICSs from records generated in the monitoring database and based on the precision requirements listed above, and as required by the methodology and guidance in the Sampling Standard.

To ensure a random selection of ICSs, random number generators were used. Each ICS in the target population is uniquely identifiable by its unique serial number. Each ICS can thus be allocated a Sample Selection Number, starting at 1 and increasing up to the total number of ICSs in the Database for that pre-defined sampling frame. Applying the random number generators, the ICS can then be randomly chosen from the defined population up to the required sample size as calculated by the Project Participant.

For parameter $\eta_{\text{new,avg},y}$, loss in efficiency of the project devices due to aging during the monitoring period will be accounted for using Standard Water Boiling Test campaigns as per AMS-II.G. v13.1, where the loss in efficiency must be monitored annually through the Standard Water Boiling Test campaign from a respective sample of the first batch.

This is not applicable for this monitoring period which is less than 1 year from the project start date, as all ICSs are newly distributed and efficiency would be assumed to be manufacturer-certified value.

For parameter $N_{j,k,y}$ visual inspection of the premises will be conducted to see if the ICS is operational and in use along with the interview with end user if required to verify that ICSs are still in use.

For parameter $\eta_{j,k,y}$, the total number of devices in operation compared to the total number distributed (according to the monitoring database) is surveyed over a representative sample. The Sampling standard shall be used for determining the sample size to achieve 90/10 confidence precision according to the latest version of Standard for sampling and surveys for CDM project activities and programme of activities.

For the parameter $BC_{b,i,y}$ 5-year measurement campaigns following the Kitchen Performance Test protocols would be carried out in non-project and project households respectively. The frequency to conduct these measurement campaigns were not due for this monitoring period, which is less than less than one year from the start date of the project.

Therefore, the quantity of fuel that would be used in the baseline scenarios were determined through the Measurement Campaign conducted by Simi Oa Consultancy Services, that is a third-party consultant team contracted by TEM along with the baseline study survey concluded in November 2022. A total of 220 houses were selected for the baseline study with 110 houses each from the Southern Highlands Province and Western Highlands Province. The selected houses were provided with an individual code from 001 to 220. Of the total number of houses, 160 were used for control cooking tests (CCT) comparing firewood consumption using ICSs ($n=80$) with traditional open fire cooking (OFC, $n=80$). At the end of the campaign, sampling challenges enabled the collection of firewood consumption data on 131 households in both

provinces, of which 60 houses used ICSs and the remaining 71 houses used traditional OFC. The results of the measurement campaign yielded a value of 5 tonnes as the firewood usage per year on a baseline stove (OFC) (file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf).

For estimating $BC_{b,i,y}$, as the average daily fuel consumption per adult equivalent parameter from the measurement campaign was obtained from 71 samples and achieved a precision value 12%, the lower bound value of **5.24 tonnes/year** is used after applying a conservativeness deduction following Section 4 of CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities, version 9.0.

For estimating $BC_{p,j,k,y}$, the average daily fuel consumption per adult equivalent parameter from measurement campaigns are used which were concluded during baseline survey in November 2022 and more recently in July 2025. The project is applying a one-off deviation for the applicability of this parameter, see Section 3.2.1. As the parameter BC_p was first measured in November 2022 and is valid for 2 years as per Section 9.2 of VM0050, v1.0, till November 2024, which is part-way through the current monitoring period. Values from the recent KPT measurement campaign concluded in July 2025 are to be used from December 2024 onwards for calculating ERs. For the ease of efficiency in calculating the ERs from this monitoring period, the higher (more conservative) value of BC_p is uniformly applied throughout the monitoring period for each respective batch. This deviation results in lower ERs, than if the BC_p values were separately applied for the monitoring period and is therefore more conservative and does not impact the applicability of the methodology, the project additionality, or the appropriateness of the baseline scenario.

For Batch 1: the value from July 2025 of **1.61 tons/year** is more conservative

For Batch 2: the value from November 2022 of **1.54 tons/year** is more conservative

Using the formulas in the section “Sample Size” below, the Project Participant will randomly sample the required number of ICSs from the primary sampling units.

2.4) Sample Size

The sample size for each monitoring activity is a subset of the total number of ICSs distributed and included in the monitoring database. The minimum sample size for each sample frame is estimated to achieve a 90/10 confidence/precision for annual and/or biennial sampling. The procedure to determine the sample of households will ensure that they adequately represent the broader project population, minimizing sampling error.

In line with sampling guidelines, all parameters can be sampled in a single survey with a random sample of households using the above-described confidence/precision levels depending on annual or biennial monitoring frequency.

As per Guidelines for Sampling and surveys for CDM project activities and programmes of activities, version 09.0 paragraph 13, there are different ways available to obtain the estimates of the parameter of interest:

1. The type of parameter of interest, that is, mean value or proportion value;
2. The target value, that is, the expected value of the parameter, which should be determined using the project participants' or the coordinating/managing entity's knowledge and experience;
3. Expected variance (or standard deviation) for that measure in the sample, based on the results from similar studies including other similar CDM project activities or previous monitoring periods, pilot studies, or from the project planner's own knowledge of the data.

For the current monitoring period, the minimum sample size for monitoring surveys was estimated by ensuring that 90/10 confidence and precision would be met for the parameter $\eta_{j,k,y}$, with an expected proportion of 0.9 using the following equation:

Table 9: Equation for sample size calculation

$n \geq \frac{1.96^2 N \times p (1-p)}{(N - 1) \times 0.1^2 \times p^2 + 1.96^2 \times p(1-p)}$		
n	=	Sample size
N	=	Population size (Total number of households/ICS)
p	=	Expected proportion

For batch 1 (N = 4,649) and batch 2 (N = 14,557), this yielded a minimum sample size of 30 and 31 samples respectively. With adequate oversampling, the number of ICSs (households) selected for each batch was fixed at 60 each, to ensure that the required level of precision is met, not only to compensate for any attrition, outliers or non-responses associated with the sample, but also to prevent a situation at the analysis stage where the required reliability is not achieved, and additional sampling efforts would be required.

Therefore, a total of 120 ICSs (households) were selected for conducting monitoring surveys. The monitoring survey was completed successfully in 96 households (n = 45 from Batch 1, and n = 51 from Batch 2).

2.5) Data Recording

The distribution record including user information was collected and recorded at the point of distribution by the staff or distributors. Alongside the unique serial number listed on the ICS, information recorded include:

- Name of ICS user or head of the household

- Address / GPS of ICS household
- Phone number of ICS user or household, where available.
- ICS model
- Date of distribution/installation
- ICS serial number
- Retailer/distributor information

The information collected from the end-user was transferred to an electronic database, which will be updated regularly. The distribution record carries all distribution information including the traditional stove type (charcoal stove/firewood stove) used prior to the ICS installation, name of representative member of household, phone number (if available and user-permitted for sharing), address etc.

A record keeping system and unique identification of the project is made possible by user's information collected through the End-User Agreement and compiled in distribution records. This will include end-user name, identification number and address/location of the user's house, ICS unique serial number and distribution date. The record keeping system will ensure that each ICS can be traced to the project activity to avoid double counting.

The Monitoring Plan applied in this project involves several key elements that ensure that the PP have high-quality, unbiased, and reliable information regarding the performance of the project in terms of implementation and outcomes, and for the purposes of calculating emissions reductions on the basis of the amount of non-renewable biomass saved by the ICSs in the project activity.

The key elements are the following:

- Project database management
- Sample Plan for the Monitoring Survey
- Data Quality, Consistency and Duplication Checks
- Monitoring Reporting

The TEM PNG's Country Manager will have overall responsibility for the implementation of the monitoring plan. The information collected will be stored in the electronic database and will be updated on ongoing basis by trained staff.

Data collection was carried out by a dedicated monitoring team of trained individuals, from TEM PNG. All technical staff responsible for installation and maintenance of the ICSs were trained in terms of the understanding the requirements on the monitoring system.

The monitoring survey was administered digitally with a questionnaire adapted from the binding questionnaire provided in the appendix of VM0050 v1.0 and other questions relevant for the monitoring of co-benefits, including photographs of the stoves, its components and the cooking

areas³⁰. The responses from the surveys were recorded on digital tablets, transferred to an electronic database and analysed to ensure that the 90/10 confidence and precision levels are met.

The monitoring survey was conducted from 16 January 2025 to 17 January 2025 where a total of 98 households were surveyed.

2.6) Organizational Structure

The project will be managed by TEM as the project participant. The roles and responsibilities are shown in Table 10 below:

Table 10: Organizational Structure and Responsibilities

Person/Entity	Responsibility
TEM Country Manager, Field Coordinator and Field Assistants	• Oversee operation of distribution centres; execution of set up activities; works with project manager on all planning. • Execution of set up activities such as recruitment and training, reporting of monitoring data. • Logistics Manager: planning; identification of target households; contractor management; overall day to day management of installation staff; weekly and monthly reporting. • Data administrators: monitoring database management; accounting; data reconciliations; monthly reporting • Pre- & Post-distribution data collection: conveying project messages; distributing the project; signing up householders wanting an ICS; sign up data capture; distribution data capture • Distribution team: management of distribution process; ensuring quality ICS distributions. • Monitoring team: gathering compliance monitoring data; gathering marketing data; data input

³⁰ Refer to Appendix V: Monitoring Survey Photographs

Person/Entity	Responsibility
TEM Director of International Projects	• Project planning and management; issue and risk management. • Oversight of management system & signoff on ICS procurement and monitoring reports, review of competencies of team members • Technical review team: technical review of process and documentation and monitoring reports. • Compliance Manager/Consultant: writing PD & monitoring reports, ensuring compliance with VCS rules.

Elaboration of Tasks:

Table 11: Tasks and Responsibilities of different roles

Person/Entity	Task	Responsibility
TEM Country Manager, Field Coordinator and Field Assistants	Manufacturing and logistics	Depending on the ICS model, complete or components for the ICSs are: • Manufactured (some imported, others produced locally) by an ICS manufacturer. • ICSs are distributed to warehouses within project region. • Coordinate the distribution of ICSs to recipient households

TEM Country Manager, Field Coordinator and Field Assistants	Household identification	• Work with local community leaders and other partners to help identify recipient households suitable for the distribution of an ICS. • Establish partnership with community leaders, NGOs and other local organizations, local partner will initiate a communication process to ensure that households understand the benefits of the ICS, that cultural issues are addressed and that users are trained in the optimal use and performance of the ICS. • TEM PNG team prior distribution will visit recipient communities in each region and ensure recipients understand and acknowledge the conditions for participation in the project. • Ensure each ICS is assigned a unique distribution number chronologically to avoid any double counting through the use of an electronic database
TEM Country Manager, Field Coordinator and Field Assistants	Distribution	• TEM PNG team train distribution teams to distribute ICSs. • Coordinate the receipt of ICSs and components in the distribution process. • TEM PNG team will be trained in the distribution of the ICS to a standardized delivery and communication procedure. • TEM PNG team will be responsible for physically distributing the ICSs to the recipient household
TEM Country Manager, Field Coordinator and Field Assistants	Data Capture	• The information collected from the TEM PNG team during the distribution of ICSs, from end-users and interactions on the monitoring phase of the project will be transferred to an electronic database,

		which will be updated regularly. The database will include a unique reference number for each ICS to avoid double counting. • TEM PNG team checks the quality of ICSs and record keeping work. If the work is satisfactory, distribution data is collected by the TEM PNG team
TEM Director of International Projects	Monitoring	Monitoring activities will be conducted as follows: • Surveys completed in the field by trained local monitoring teams. • Data captured by the monitoring teams is passed to the TEM Head of Science and Sustainability • Data is checked for completeness, consistency, and accuracy. • Project manager summarizes data in a report to TEM Head of Science and Sustainability • Writing of monitoring reports for each monitoring period. • Technical review by in-house technical team

2.7) QA/QC

In order to minimize sampling errors, such as non-responses and errors, over-sampling from the population was carried out to ensure a total number of respondents that meets the required level of precision. This will ensure the integrity of the sample group and maintains the randomness of participant selection.

As the required level of precision was achieved, not further adjustments were required to be made to the sampling plan or monitoring surveys.

24 households out of the 120 selected for surveying could not be visited during the field sampling dates of 16-17 January 2025, due to the members of the household not available at the time of the visit by the field team, or exceptional cases related to death of kin. Where a

survey may not be completed, or where there is a non-response, the reasons are clearly documented.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

Baseline emissions are calculated as follows:

$$BE_y = \sum_{i,j,k} EC_{i,y} \times N_{j,k,y} \times n_{j,k,y} \times (EF_{b,i,CO_2} \times f_{NRB,y} + EF_{b,i,nonCO_2}) \quad (1)$$

Where:

BE_y	= Baseline emissions during year y (t CO ₂ e)
$EC_{i,y}$	= Average energy consumption of baseline device type i in year y (TJ)
$N_{j,k,y}$	= Number of commissioned project devices of type j from batch k in year y
$n_{j,k,y}$	= Proportion of commissioned project devices of type j from batch k that remain operating in year y (fraction)
EF_{b,i,CO_2}	= CO ₂ emission factor for fuel used by baseline device type i in the baseline scenario (t CO ₂ /TJ)
$f_{NRB,y}$	= Fraction of woody biomass that is established to be non-renewable used by baseline device in year y ; this variable is not considered for fossil fuels (fraction)
$EF_{b,i,nonCO_2}$	= Non-CO ₂ emission factor for fuel used by baseline device type i in the baseline scenario (t CO ₂ e/TJ)
i	= Baseline device type and its respective fuel type
j	= Project device type and its respective fuel type

Average Energy Consumption of Baseline Device ($EC_{i,y}$)

The average energy consumption of baseline device type i is calculated as follows:

$$EC_{i,y} = BC_{b,i,y} \times NCV_{b,i} \quad (2)$$

Where:

$BC_{b,i,y}$	= Fuel used per baseline device type i during year y (tonnes) ¹⁷
$NCV_{b,i}$	= Net calorific value of baseline fuel for baseline device type i (TJ/tonne)

The quantity of fuel that would be used in the baseline scenario was determined through [Option 1: Measurement Campaign](#) conducted by Simi Oa Consultancy Services that is a third-party

consultant team contracted by TEM along with the baseline study survey concluded in November 2022. A total of 220 houses were selected for the baseline study with 110 houses each from the Southern Highlands Province and Western Highlands Province. The selected houses were provided with an individual code from 001 to 220. Of the total number of houses, 160 were used for control cooking tests (CCT) comparing firewood consumption using ICSs (n=80) with traditional open fire cooking (OFC, n=80). At the end of the campaign, sampling challenges enabled the collection of firewood consumption data on 131 households in both provinces, of which 60 houses used ICSs and the remaining 71 houses used traditional OFC. The results of the measurement campaign yielded a value of 5.97 tonnes as the firewood usage per year on a baseline stove (OFC) (file titled: TEM 2025_Cookstoves Highlands region report_v3.pdf).

As the average daily fuel consumption per adult equivalent parameter from the measurement campaign was obtained from 71 samples and achieved a precision value 12.36%, the lower bound value of 5.24 tonnes/year is used after applying a conservativeness deduction following Section 4 of CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities, version 9.0 as shown in *Table 12*.

Table 12: Conservativeness deduction for Fuel consumption (November 2022)

	Baseline (OFC)	Project (ICS)
Avg adult consumption/year (tonnes)	1.23	0.28
Standard Deviation	0.77	0.17
Precision	12.36%	12.98%
Avg Household size	4.85	
Avg consumption/ household /year (tonnes)	5.97	1.38
Upper bound	6.71	1.54
Lower bound	5.24	1.18
After conservativeness correction		
Fuel consumption/household/year	5.24 (BC _b)	1.54 (BC _p)

Stove stacking

It was observed from the monitoring surveys that some households ((16% in Batch 1 and 8% in Batch 2) were using the baseline method of open fire cooking (OFC) for a part of the meals cooked in conjunction with the ICSs distributed as part of the project activity. To be conservative in estimating the GHG emissions reductions, the usage rates for each batch is discounted

correspondingly to account for stacking in the project scenario, up to a cap of 90%. This is detailed in Section 5.4 of this document.

Baseline Emission for Project Instance:

Table 13: Determining Average Energy Consumption of Baseline Device ($EC_{i,y}$)

Equation 2	$EC_{i,y} = BC_{b,i,y} \times NCV_{b,i}$		
	$BC_{b,i,y}$	$NCV_{b,i}$	$EC_{i,y}$
	5.24	0.02	0.08

Table 14: Number of Cookstoves in each batch ($N_{j,k,y}$)

$N_{j,1,y}$ (Batch 1)	$N_{j,2,y}$ (Batch 2)
4,649	14,557

Therefore, baseline emissions will be calculated by batch as follows:

Table 15: Calculation of baseline emissions for each batch (BE_y)

Equation 1	$BE_y = EC_{i,y} \times N_{j,k,y} \times n_{j,k,y} \times (EF_{b,i,CO_2} \times f_{NRB,y} + EF_{b,i,nonCO_2})$						
	$EC_{i,y}$	$N_{j,k,y}$	$n_{j,k,y}$	EF_{b,i,CO_2}	$f_{NRB,y}$	$EF_{b,i,nonCO_2}$	BE_y
Batch 1	0.08	4,649	0.84	112	0.60	9.46	6,225
Batch 2	0.08	14,557	0.90	112	0.60	9.46	13,923

5.2 Project Emissions

Project emissions are calculated as follows:

$$PE_y = PE_{energy,y} + PE_{others,y} \quad (6)$$

Where:

- PE_y = Project emissions during year y (t CO₂e)
- $PE_{energy,y}$ = Project emissions from energy consumption of project devices in year y (t CO₂e)
- $PE_{others,y}$ = Project emissions from other sources in year y (t CO₂e)

$PE_{energy,y}$ from Biomass is determined as follows:

$$PE_{energy,y} = \sum_j \sum_k BC_{p,j,k,y} \times N_{j,k,y} \times NCV_{p,j} \times n_{j,k,y} \times (EF_{p,j,CO_2} \times f_{NRB,y} + EF_{p,j,nonCO_2}) \quad (7)$$

Where:

- EF_{p,j,CO_2} = CO₂ emission factor for fuel used by project device type j in the project scenario (t CO₂/TJ)
- $EF_{p,j,nonCO_2}$ = Non-CO₂ emission factor for fuel used by project device type j (t CO₂e/TJ)

For estimating $BC_{p,j,k,y}$, the average daily fuel consumption per adult equivalent parameter from measurement campaigns are used which were concluded during baseline survey in November 2022 and more recently in July 2025. The project is applying a one-off deviation for the applicability of this parameter, see Section 3.2.1. As the parameter BC_p was first measured in November 2022 and is valid for 2 years as per Section 9.2 of VM0050, v1.0, till November 2024, which is part-way through the current monitoring period. Values from the recent KPT measurement campaign concluded in July 2025 are to be used from December 2024 onwards for calculating ERs. For the ease of efficiency in calculating the ERs from this monitoring period, the higher (more conservative) value of BC_p is uniformly applied throughout the monitoring period for each respective batch. This deviation results in lower ERs, than if the BC_p values were separately applied for the monitoring period and is therefore more conservative and does not impact the applicability of the methodology, the project additionality, or the appropriateness of the baseline scenario.

In November 2022, the average daily fuel consumption per adult equivalent parameter from the measurement campaign was obtained from 60 samples and achieved a precision value 12%, the upper bound value of 1.54 tonnes/year is used after applying a conservativeness deduction as per Section 4 of CDM Standard for Sampling and Surveys for CDM Project Activities and Programmes of Activities, version 9.0.

For the July 2025 KPTs, the measurement campaign yielded an average consumption of 1.61 tonnes/year per device from 50 samples in Batch 1, 1.49 tonnes/year from 42 samples in Batch

2, with a precision value of 7.45% and 8.23% respectively. As the precision values are under 10%, no conservativeness deductions were necessary.

For Batch 1 ($BC_{p,j,1,y}$): the value from July 2025 of **1.61 tons/year** is more conservative

For Batch 2 ($BC_{p,j,2,y}$): the value from November 2022 of **1.54 tons/year** is more conservative

$PE_{others,y}$ is determined as follows:

$$PE_{others,y} = PE_{transp,y} + PE_{prod,y} + PE_{fugitive,y} + PE_{backup,y} \quad (9)$$

Where:

$PE_{transp,y}$ = Project emissions due to fuel transportation in year y (t CO₂e)

$PE_{prod,y}$ = Project emissions due to fuel production in year y (t CO₂e)

$PE_{fugitive,y}$ = Fugitive emissions in year y (t CO₂e)

$PE_{backup,y}$ = Project emissions from backup generators in year y (t CO₂e)

As the project design and boundary does not involve fuel transport, production, fugitive emissions or emissions from backup generators, this calculated parameter does not apply to the project activity.

Project Emission for Project Instance:

Table 16: Calculation of Project Emissions for each batch ($PE_{energy,y}$)

Eq 7	$PE_{energy,y} = BC_{p,j,k,y} \times N_{j,k,y} \times NCV_{p,j} \times n_{j,k,y} \times (EF_{b,i,CO_2} \times f_{NRB,y} + EF_{b,i,nonCO_2})$							
	$BC_{p,j,k,y}$	$N_{j,k,y}$	$NCV_{p,j}$	$n_{j,k,y}$	EF_{b,i,CO_2}	$f_{NRB,y}$	$EF_{b,i,nonCO_2}$	$PE_{energy,y}$
Batch 1	1.61	4,649	0.0156	0.84	112	0.60	9.46	1,909
Batch 2	1.54	14,557	0.0156	0.90	112	0.60	9.46	4,090

5.3 Leakage Emissions

As per VM0050 v1.0, Leakage emissions (LE_y) are accounted as follows:

Leakage emissions associated with fossil fuel use:

- Increased emissions from fossil fuel use by non-project participants

An adjustment factor of 0.95 is applied to the GHG emission reductions. (i.e., $0.95 \times (BE_y - PE_y)$). This results in 712 t CO₂e for this monitoring period.

As the project activity does not associate with renewable biomass ($LE_{RB,y}$), $LE_{RB,y}$ is '0'.

5.4 GHG Emission Reductions and Carbon Dioxide Removals

Net GHG emission reductions are calculated as follows:

$$ER_y = (BE_y - PE_y) \times 0.95 - LE_{RB,y} \quad (11)$$

Where:

ER_y = Emission reductions during year y (t CO₂e)

$LE_{RB,y}$ = Leakage emissions associated with use of renewable biomass during year y (t CO₂e)

The emissions reductions for the crediting period of the project activity in each batch are calculated per Equation 11 in *Table 17*.

Table 17: Calculation of Emission Reductions

Equation 11	$ER_y = (BE_y - PE_y) \times 0.95 - LE_{RB,y}$			
	BE_y	PE_y	$LE_{RB,y}$	ER_y
Batch 1	6,226	1,909	216	4,101
Batch 2	13,924	4,090	492	9,342
Total	20,150	5,999	708	13,443

Vintage period	Baseline emissions (tCO ₂ e)	Project emissions (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Reduction VCUs (tCO ₂ e)	Removal VCUs (tCO ₂ e)	Total VCUs (tCO ₂ e)
23-Sep-2024 to 31-Dec-2024	20,150	5,999	712	13,443	0	13,443
Total	20,150	5,999	712	13,443	0	13,443

Vintage period	Ex-ante estimated reductions/removals	Achieved reductions/removals	Percent difference (%)	Explanation for the difference
23-Sep-2024 to 31-Dec-2024	103,526	13,443	-87	The ex-ante estimates assumed a distribution of 100,000 ICSs, while during this monitoring period (phase 1 of distribution) the actual number of ICs distributed were 19,206.

Total	103,526	13,443	-87	
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APPENDIX I: COMMERCIALY SENSITIVE INFORMATION

No commercially sensitive information has been excluded from the public version of the project description.

APPENDIX II: LSC OUTREACH

1. Newspaper Advertisements:

There were two newspapers published nationally in PNG. As the project activity would be a grouped project activity, TEM saw the importance of announcing the to the public and informing stakeholders of the project, as well as briefing them on the upcoming local stakeholder consultations happening in the Highlands Provinces. The following table shows the date of when these advertisements were published in the newspaper:

S/N	Publishing House	Advertisement Published on: (date)	Supporting Documentation
1	Post Courier Newspaper	14/02/2023	Invoices from respective newspapers for publishing, picture of advertisement in newspapers

2. Letters of Notification to Provincial Governments:

Letters of Notification were also hand delivered to provincial government administrative assistants to brief them on the project activity as well as the consequent co-benefits to their local community that came with the project. Together with the Letter of Notification, a project brief was also attached to aid their understanding. The following table shows the dates of when the letters of no-objection were received from the provincial governments:

S/N	Province	Document Received on: (date)	Supporting Documentation
1	Western Highlands Province	13/02/2023	Letter of No-Objection
2	Jiwaka Province	15/02/2023	
3	Eastern Highlands Province	16/02/2023	
4	Southern Highlands Province	08/03/2023	

During TEM's consultations with the provincial governments, the administrative assistants expressed a high level of interest in the project and were supportive of its implementation.

3. Radio Broadcasts

To further TEM's outreach into the Highlands Provinces regarding the LSCs, radio broadcasts were utilized. All radio broadcasts were done by the National Broadcasting Commission which then broadcasted TEM's announcements to individual provinces at a frequency of 1 time a day, over 10 days. Each province-level broadcast would then inform the specific province of the date, location and time of

which the LSC was happening within their area. The following table shows the start and end dates of these broadcasts:

S/N	Province	Start of Broadcast: (date)	End of Broadcast: (date)
1	Western Highlands Province	29/02/2023	12/03/2023
2	Jiwaka Province	29/02/2023	12/03/2023

The radio broadcast could only be done for 2 highlands provinces as it was found that the radio stations for the remaining two provinces had been shut down.

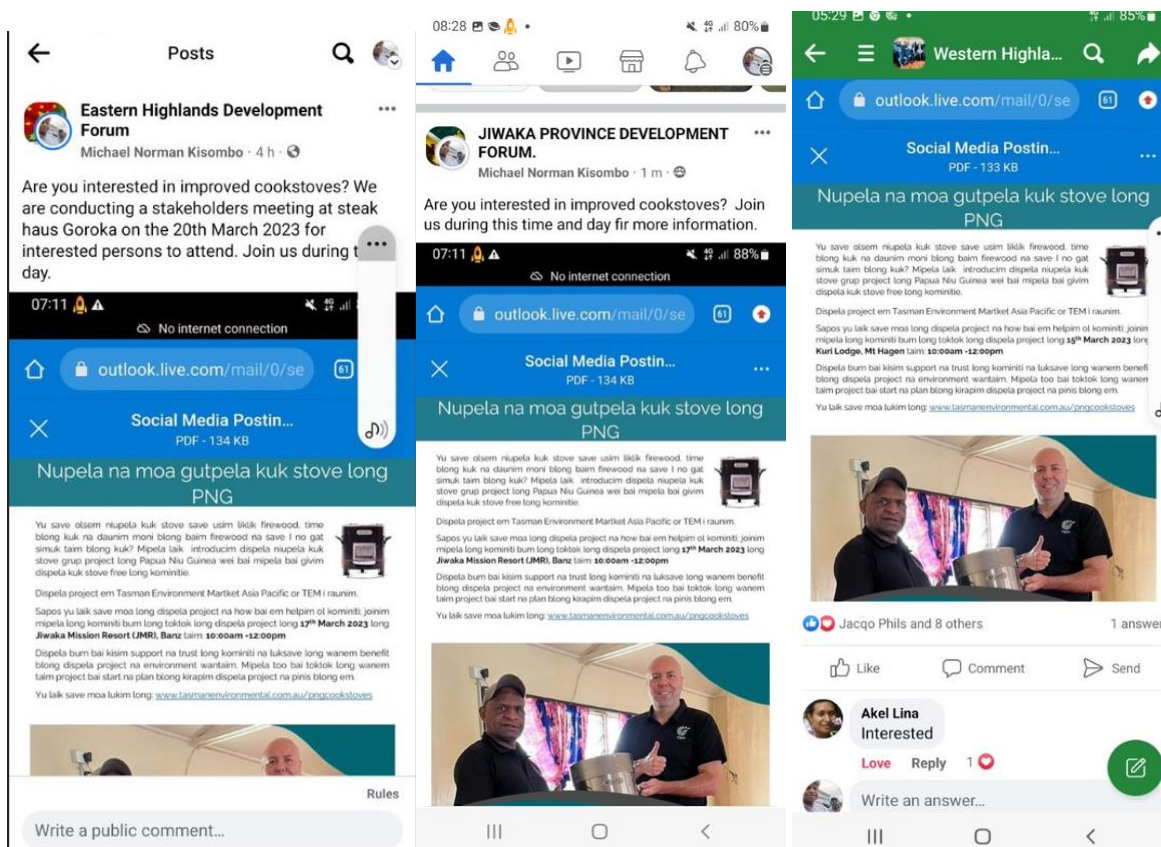
4. Invitation to Churches

TEM is aware that churches and church leaders play an important role in the local community within PNG. As such, invitations to churches were also hand delivered to churches within each province to invite church leaders and elders to participate in the LSC, as well as spread the word about the importance of the LSC within their community by word-of-mouth. The following table shows the dates these invitations were delivered to churches within the provinces:

S/N	Province	Delivery of Invitation: (date)	Supporting Documentation
1	Southern Highlands Province	10/02/2023	Acknowledgement of Receipt by Church Bodies: SHP
2	Western Highlands Province	15/03/2023	Acknowledgement of Receipt by Church Bodies: WHP, Jiwaka, EHP
3	Jiwaka Province	17/03/2023	
4	Eastern Highlands Province	20/03/2023	

5. Social Media Postings

TEM also understood that the majority of the stakeholders received news of events happening within their area through social media. For this, flyers depicting the details of the LSC, and introduction of the project were developed and posted on each province's dedicated Facebook development forums to further boost stakeholder awareness of the LSC. Postings of these flyers started on 13/03/2023.

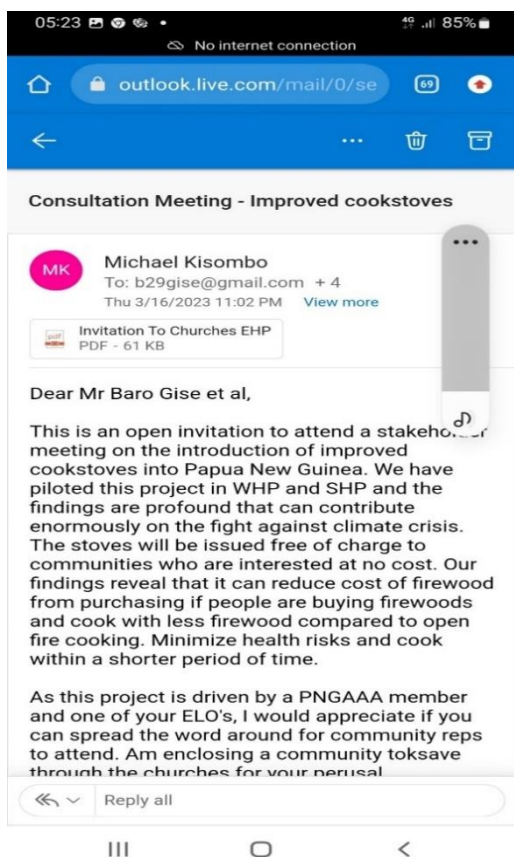
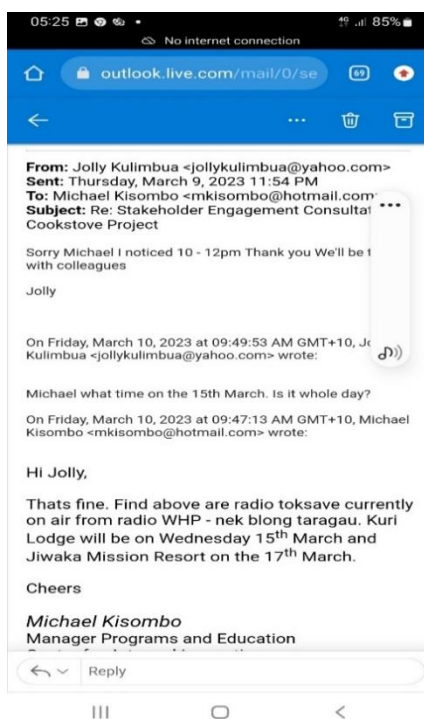
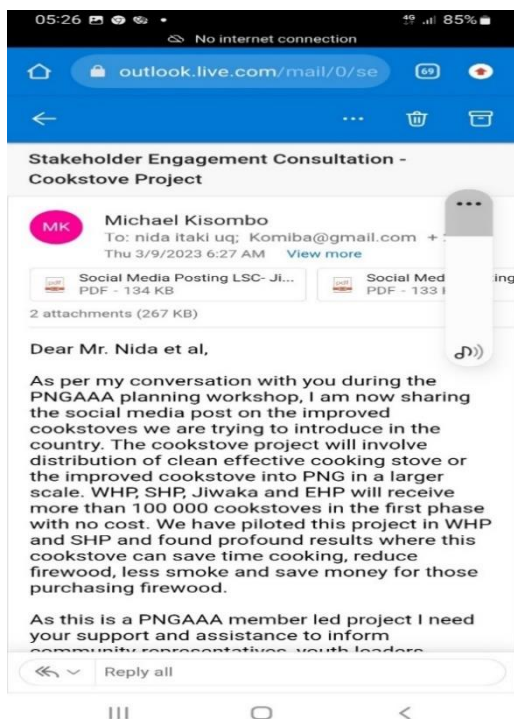


6. Email to PNG Australia Alumni Association (PNGAAA) members

From within TEM's local partner's connections, TEM also reached out to members from WHP and EHP via email. The table below shows the dates these emails were sent out:

S/N	Province	Date of Email Sent: (date)
1	WHP	09/03/2023
2	EHP	16/03/2023

Screenshots of the emails sent out can also be found below:



APPENDIX III: STAKEHOLDER FEEDBACK

1. Southern Highlands Province:

Questions:

S/N	Question	Answer
1	Ms Julie Ponaria (a teacher from SHP) enquired that there are many big countries contributing to the destruction of the ozone layer and now trying to give (us) cookstoves to stop carbon emissions. She asked if there is a story behind this small cookstove?	Yes, there are many industrial countries that are destroying the ozone layer because they have big factories and industries. We (people of PNG) also do so when cutting down trees for farming and cooking etc... This project is to reduce carbon emissions by using less firewood for daily needs. Hence, saving trees that consume carbon and restore them. That is the story behind this project.
2	Ps Joseph Yapasi of Tente Ward 2 commented that 50 000 cookstoves is not sufficient to cater for all households in Southern Highlands and there may be risks when distribution. Therefore, he suggested that distribution be more targeted. E.g distribute through church networks or distribute to a particular district first and later target other districts.	It was noted for TEM to decide.
3	Ms Regina Kamerepa pointed out that since the cookstoves are going to be issued free, people are likely to rush to get one. Hence, can also steal or pull them from the distributors during the distribution. And asked how TEM would manage that scenario?	This is a threat we are aware of and are working on a strategy on how this can be mitigated during distribution. Currently, it is planned that security will be present during the distribution process. End-users will also be advised to keep their cookstoves in a safe place to deter theft of the device.

Reflections:

S/N	Reflections
1	A youth rep commented that the level of presentation was too technical and high given the understanding level of the communities and stakeholders present asked if the presentation can be more tailored to their understanding level.

2	A youth rep commented that the level of presentation was too technical and high given the understanding level of the communities and stakeholders present asked if the presentation can be more tailored to their understanding level.
3	A youth rep commented that the level of presentation was too technical and high given the understanding level of the communities and stakeholders present asked if the presentation can be more tailored to their understanding level.
4	Mr. Peter Wari who is in charge of climate change in the province acknowledged the project as it was really going to benefit the community. He further stressed about the importance of the project activity and encouraged the stakeholders to support the project.
5	Most stakeholders mentioned that there is an increase in the usage of firewood and most of their trees (caussarina/yar) are diminishing because people are either chopping them to sell as there is a high demand or for household use. Therefore, the introduction of such cookstoves will be relieving for them.
6	Most stakeholders mentioned that there is an increase in the usage of firewood and most of their trees (caussarina/yar) are diminishing because people are either chopping them to sell as there is a high demand or for household use. Therefore, the introduction of such cookstoves will be relieving for them.
7	Another youth mentioned that he provides firewood for three separate houses as most are windows who have no husbands. With the new cookstove, he believes will relieve him from time and struggles he is facing foraging for firewood for all he's mothers.
8	A Health worker emphasized that inhaling of smoke using normal open fire cooking is contributing to a lot of respiratory illnesses and with the introduction of the ICS she believes will minimize such especially mothers and girls.
9	Besides the above reflections and questions, there were quiet a good number of people giving positive comments and supported the project activity. They even asked to increase the volume from 50 000 ICSs to enough to cater for all households the province.

2. Western Highlands Province:

Questions:

S/N	Question	Answer
1	Mr. Frank Kuri Cr from Mul Baiyer asked, since it's going to affect the rural communities, can the councilors get involved?	Yes, Councilors will get involved to identify heads of households but we do not want this project to be politicized, either be councilors or local MPs.
2	Mr. Nida L Taki, a lecturer at the Agriculture College, Mt Hagen asked if there are further studies made to determine the level of smoke that is given out during normal open fire cooking to see how much carbon emissions are produced.	That's an interesting question. However, this project used observation method to observe the level of smoke when cooking both

		during the control cooking test and open fire cooking.
3	Mr Dickson Wia stressed that firewood is not only used for cooking in the highlands but for other purposes too. Will this cookstove provide for all?	Yes, we use firewood for many other purposes like for heat, light, storytelling etc. This cookstove is for cooking and it can also heat up the house.
4	Rex Parak enquired if the design of the cookstove can be made locally?	The brand and model of cookstove was chosen specifically for its high efficiency. TEM will consider a local brand in the future if it fits the specifications required to yield the highest efficiency in the combustion of firewood.

Reflections:

S/N	Reflections
1	The District Administrator for Dei District thanked the team for introducing this new innovative idea of cooking as people in his electorate are running out of firewood. He welcomed the project initiative and asked to include all rural communities in his electorate.
2	Those who came from the urban centers also requested that they should be given priority as they are using a lot of money to purchase firewood and its already becoming a burden.
3	One of the participants mentioned that the project is very beneficial as it saves time foraging for firewood and a financial relief in buying firewood.
4	A youth mentioned that this project will benefit the youth as he sees that firewood are already running out.
5	Another supported by saying, the initiative taken by TEM is indeed very vital to reduce carbon monoxide as well as relieving economic burden and reduce health related issues. She suggested that it would be good to involve the councilors.
6	One said, this project must not stop after 10 years. It must continue till the world ends.

7	A health worker supported the project by saying, as a health worker this project if rolled out effectively will help reduce the number of sicknesses like pneumonia, chronic obstructive lung disease and eye diseases as well.
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3. Jiwaka Province:

Questions:

S/N	Question	Answer
1	Mr. Lumun Mingal asked how can PNG be compensated well since we are less carbon emitters yet are asked to further reduce the emissions through such cookstove project?	This cookstove project aims to contribute to the wellbeing of the stakeholders in PNG by lessening the burden of purchasing or collecting firewood and improving the respiratory health of end-users exposed to hazardous smoke from the burning of firewood. Cookstoves being distributed for free is paid for through carbon finance and is one method of compensating stakeholders using the carbon credit scheme.

Reflections:

S/N	Reflections
1	A local landowner thanked TEM for this initiative and welcomed everyone to JMR. Having said that he suggested that this project be based there as it is the model village for Jiwaka and must be made the main distribution point.
2	Another said, there are many NGOs like IOM, UNICEF, etc. who come and deliver similar community-based projects but we the people seem to sell those items. He was worried the cookstoves might be sold in a similar manner and appealed to other participants to discourage such practices.
3	A female teacher mentioned that as a teacher she moves from one area to another with kerosine stoves and with the introduction of the improved cooking stove, hers and other teachers would benefit greatly. Not only that but emphasized that they usually ask their students to bring firewood for them once a week and this project will ease that burden.

4	Another person suggested that every household in Jiwaka must receive a cookstove as they are facing difficulties in looking for firewood.
5	Sammy Buka of Banz mentioned that the project is a bottom up where local communities will benefit. He said it's a sickness free project.
6	There were other supportive comments like, TEM must continue this project to other provinces in PNG, delivered through churches, schools and LLG wards.
7	When asked if there are risks involved like those of you selling firewood, would this project affect you? Everyone said no, not at all as they see there is a lot of benefit to the rural communities.
8.	Someone also wanted more than 50 000 cookstoves in Jiwaka as they believe they are good people.

4. Eastern Highlands Province:

Questions:

S/N	Question	Answer
1	Mr Rex Yagusa mentioned that there is a high possibility of the cookstoves being stolen by thieves or destroyed by natural and man-made disasters and asked in such a case, how can we register these complains or cases?	Currently, it is planned that security will be present during the distribution process. End-users will also be advised to keep their cookstoves in a safe place to deter theft of the device.

Reflections:

S/N	Reflections
1	One of the ladies (video graphed) mentioned that this cookstove will lessen the burden for those mothers in the rural setting. She said if they don't sell, there is no money to purchase firewood so they would use plastics and cardboard to cook. This project will really relieve those struggling mothers living in the urban areas.
2	Another suggested having the distribution point centralized and allowing only genuine households to collect. Later field officers can be sent to their locations for registration such as GPS points etc.

3	Someone in the group mentioned that economic benefits are huge as they will be working as the project team to distribute.
4	Those who attended expressed support for the project and demanded if more is distributed in EHP given their population density.
5	No risks or harm mentioned when implementing this project.
6	One participant mentioned that tree-cutting is happening almost every day. More supplies to meet population will help reduce carbon emission.

APPENDIX IV: STAKEHOLDER CONSULTATIONS

During the current monitoring period, the following stakeholder interactions in SHP were conducted:

Ongoing consultation	Non-formal education and training was provided to local people to be converted to local staff
Date(s) of stakeholder consultation	08 July 2024 – 11 July 2024
Communication of monitored results	Training was provided by TEM staff in the local language. Training modules included: 1) Introduction to ICS Community Engagement & Communication 2) ICS Health & Safety Field Team Roles 3) Data Collection, tools, and Hands on Practice 4) Health, Safety and Security for the Project An informal quiz was conducted at the end of the training course to test the local staff's knowledge. The compulsory passing rate of the quiz was at least 70%. The passing rate of all training workshop participants at the end of this training workshop was 100%.
Consultation records	Training records of all training courses were kept, and certificates were presented to staff who had completed their training.
Stakeholder input	All staff trained were briefed that they could always seek clarification from the TEM PNG office in the event that they had any questions on training materials or company policies. Any input on training received during the training process will be re-evaluated if needed.
Ongoing consultation	Awareness and Household Registration of houses to take part in the ICS program.

Date(s) of stakeholder consultation	19 August 2024 – 31 October 2024
Communication of monitored results	Prior to the distribution of ICSs, awareness was conducted on the benefits of ICSs in every ward and village. Households interested in taking part in the project registered through TEM's field assistants using their particulars and household information. Households who met the household selection criteria were onboarded into the project and were given a token flyer to exchange for an ICS.
Consultation records	Information was recorded in documents by software to keep track of the ownership of the ICS. A total of 19,206 household registrations were done.
Stakeholder input	Interested households provided information such as its geographical location and contact details of the head of the family or end-user. Consent form/Household Registration form were also signed off by the head of the family or end-users.

Ongoing consultation	Collection of an ICS by end-users who have undergone Household Registration previously
Date(s) of stakeholder consultation	23 September 2024 – 31 October 2024
Communication of monitored results	The head of the family or end-users brought their token flyers to the distribution sites where they were checked and collected by the distribution team. Collected tokens were exchanged for an ICS.
Consultation records	A data entry officer was present to manage data collected by the distribution team. A total of 19,206 tokens flyers were collected and exchanged for an ICS.
Stakeholder input	All information provided was collected and managed by the data entry officer and the warehouse and logistic officer

	who manages the data collected from the households. Information regarding end-users would be updated once they have received the ICS. The information collected will include: • Name of ICS user, • Address/GPS of ICS household • Phone number of ICS user • ICS model • Date of distribution and installation • Name of staff that delivered and installed the ICS • Photographic Evidence
Ongoing consultation	A day of training was conducted for TEM's in-country field assistants before conducting the monitoring survey. This was followed by two days of ground monitoring surveys, conducted for this monitoring period's end-users as per the sampling plan.
Date(s) of stakeholder consultation	15-Jan-2025 to 17-Jan-2025
Communication of monitored results	Training workshops were conducted in Tok Pidgin, the native language of Papua New Guinea, by TEM's in-country team. During this workshop, field assistants were briefed on the logistics and importance of this survey.
Consultation records	Results from the ground monitoring survey were collected and collated to be used in the ER calculation. A total of 120 households were selected, and 96 Sampling Surveys were successfully completed.
Stakeholder input	All inputs which were required by the sampling plan was collected from end-users through measurements, observations and a list of survey questions.

APPENDIX V: MONITORING SURVEY PHOTOGRAPHS

The photographs below show the Improved Cookstoves, its components, and the cooking areas within the participating household:



Appendix Figure1. A participant takes a picture with her cookstove while a field assistant conducts the monitoring survey with her.



Appendix figure 2. A participant takes a picture with his cookstove outside his home. Obvious soot and dust marks can be seen on the cookstove indicating usage.